

BC State-Space Computer

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Abstract

That excitations are countable, that energy is additive, and that independent composition multiplies are three laboratory-level basic facts; their minimal algebraic form is the Bost–Connes (BC) covariance structure [3]. Starting from this we define the BC state-space computer: a controllable physical system that carries the BC covariance relations in finite truncation within finite experimental boundaries. A run is the complete chain that takes external conditions through boundary formation, preparation, constrained evolution, stopping and readout, to return coupling; computation is defined on the run law by a task, a reusable boundary program, physical compilation, and completion criteria.

Three structural dilemmas of the prevailing paradigm close at the physical-structure level of this machine. Content processing is carried directly by the state space, and step-by-step replay does not arise; statistics and thermalization are native resources, and pseudorandom reconstruction does not arise; the interface to the world is carried by the physical coupling of transduction and actuation, and symbolic replicas do not arise. The supporting construction consists of the arithmetic workspace (prime occupations carry multiplicative composition, phase sectors carry addition), machine parts organized into an arithmetic and an interaction module, and running instances that progress from elementary operations to sustained tracking. The thermalization mechanism is stated conditionally, all verdict quantities are pre-registered by a numerical twin, and whether devices physically realize them is left to independent experiments.

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1 Definition of the BC State-Space Computer

Every computer is a physical system; the states it can reach, together with their relations of composition, evolution, and statistics, constitute its state space. The capability boundary of a computer is set by the relation between its state space and its computational process. The prevailing paradigm lets symbols carry computation: physical states and processes are first compressed into symbolic descriptions, and dynamics, statistics, and environmental coupling are then reconstructed at the symbolic layer; the overhead of this reconstruction constitutes its dilemmas.

1.1 Dilemmas of the Prevailing Paradigm

Digital computers and quantum circuits impose the same arrangement on their state spaces: only a few latchable symbol states are used, spontaneous evolution is suppressed into holding, and the process is driven from outside. Three dilemmas follow.

Dilemma one: step-by-step replay of dynamics. A digital computer compresses physical processes into symbolic ones: the degrees of freedom carrying the bits are engineered to hold

and not to evolve, bits are maintained by latching and refresh, and state changes are driven by external instructions. This structure buys universality at the price of leaving its own state space idle: any structure can be described symbolically. The dynamics of the target system then has nowhere to live and must be replayed step by step by instructions. The resources the replay requires are set by the state space of the target system: a physical system evolves along its own dynamics and needs no describing apparatus; a symbolic machine replaying the same evolution must store and update a complete state description. For the state space $\mathcal{H}$ of n two-level degrees of freedom,

$$\dim \mathcal{H} = 2^n, \quad n = 50 : \quad 2^{50} \approx 1.1 \times 10^{15} \text{ complex amplitudes} \approx 18 \text{ PB},$$

exceeding the memory of any existing supercomputer and doubling with every added degree of freedom; one step of evolution must moreover act once on that dimension. This is the quantitative form of the exponential growth, with entanglement scale, of the classical cost of simulating quantum systems [1]; first-principles molecular simulation at chemical accuracy has therefore remained confined to small molecules. Quantum circuits retain superposition and entanglement, but the arrangement of the state space is unchanged: qubits are decoupled to nearly zero spontaneous dynamics, coherence protection amounting to frozen evolution; the process is driven gate by gate by external pulses, and environmental coupling is treated as error, isolated by massive physical-qubit redundancy. The exponential dimension is kept; evolution is still replaced by driving.

Dilemma two: reconstruction of statistics. A symbolic machine has no native probability or temperature: measurement statistics are imitated by pseudorandom generators, thermal equilibrium is approached step by step by sampling algorithms, and sample counts and mixing times constitute the main overhead. Photorealistic rendering is one example: in the physical world light propagates along all paths simultaneously and superposes by itself, while a symbolic machine samples light paths one at a time by Monte Carlo, millions of pixels per frame and hundreds of samples per pixel; equilibrium computations in statistical physics are likewise governed by the mixing times of Markov chains. Statistics that physical processes carry natively must be reconstructed sample by sample at the symbolic layer.

Dilemma three: the symbolic replica of the world. Tasks closed in a loop with the environment must maintain a replica of the world at the symbolic layer and synchronize it in real time: the outside world is first compressed into symbols, the machine decides on the replica, and actuators then return the decision to the physical world. The human brain accomplishes whole-body control, perception, and learning at about 20 W; humanoid robots re-solve motion optimization every millisecond at roughly two orders of magnitude more power, because balance and gait are not carried by their structure. The unpowered passive walker [30] walks down a slope by pure mechanics, its gait an eigenmode of the structure; this dilemma disappears wholesale once structure carries the dynamics. Open environments further make the replica unbounded: after years of investment, autonomous driving still operates inside geofenced areas, the synchronization rate of the symbolic replica falling below the world's own bandwidth of evolution.

The three dilemmas share one root: when structure is not carried by the state space, it must be rebuilt in symbols; a new paradigm must therefore start from the state space.

1.2 Requirements for the New Paradigm

Each dilemma points to a structure absent from the state space. A machine that avoids them must satisfy five requirements.

- (1) Combinatorial structure is given by the state space: many-body composition rules come from physical composition, not from conventions of cells and gate sets; exponential dimension is native capacity, no longer replayed step by step.
- (2) Dynamics is carried by the state space: evolution is undertaken spontaneously by the system, and the program only imposes constraints; step-by-step replay is thereby eliminated.
- (3) Statistics is native: measurement probabilities and thermalization are directly usable resources; pseudorandom numbers and sampling reconstruction are thereby eliminated.
- (4) The environment is a participant: the outside world enters the run through physical boundaries, and results return through physical coupling; perception and actuation do not pass through symbolic replicas.
- (5) Native dynamics coexists with algebraic exactness: analog computers possessed spontaneous dynamics and were eliminated by noise accumulation and channel crosstalk; the state space of the new paradigm must hold continuous evolution and an exact combinatorial algebra at once.

We argue below, item by item, that the BC state space satisfies these five requirements.

1.3 The BC State-Space Computer

Reference [3] argues, from three laboratory-level basic facts, for a unified structure of the physical world: that excitations are countable, energy is additive, and independent composition multiplies uniquely determines $(\mathbb{N}, +, \times)$. This paper takes that claim as a premise; the paradigm's own thesis, together with its adjudication clause, is stated at the end of §3.12. The minimal algebraic form of the three facts is the BC covariance structure, whose state space yields in turn the axioms of quantum mechanics, four-dimensional spacetime, and the spectrum of matter and interactions, and, under channel identification, the observed numbers of the Standard Model with zero free parameters.

The physical world is organized by this state space; the state space that carries computation is therefore taken to be this same structure. The machine shares the world's mode of organization and is a controllable realization of this structure; satisfaction of the five requirements is verified item by item as follows.

Requirement (1) is met by unique factorization: the state space is organized as a tensor product along prime directions, many-body composition rules are given by arithmetic, and no cell convention is needed; exponential dimension is the native capacity of this tensor structure.

Requirement (2) is met by the modular flow: the energy $H = \log n$ generates time evolution, and thermal contact with a KMS environment provides relaxation; evolution and thermalization are carried spontaneously by the system, and the program need only choose boundaries and constraints.

Requirement (3) is met by the KMS structure: the Born form of measurement probability is derived from KMS states and modular structure [3]; statistics and temperature are native readouts of the machine, requiring no reconstruction.

Requirement (4) is met by boundary coupling: external conditions enter the run through transduction, and results are sent back to the environment through return coupling; perception and actuation are carried by physical coupling, without passing through symbolic replicas.

Requirement (5) is met by the rigidity of arithmetic: occupations take an integer spectrum, so rounding does not exist; mode frequencies take $\{\log p\}$, and unique factorization excludes all spurious resonance channels at the ideal algebraic level; nearly degenerate channels at finite linewidth are a matter of implementation precision (§8.1). Continuous evolution coexists with exact algebra, and the failure mode of analog computers is structurally closed off.

Taken together, the five requirements characterize a way of computing. **State-space computing** is the paradigm in which a physical state space and its intrinsic dynamics carry computation: states carry data (requirement (1)), spontaneous evolution carries process (requirement (2)), measurement statistics carry results (requirement (3)), and programs are constraint-imposing boundary conditions (requirement (4)); requirement (5) restricts which state spaces can carry the paradigm: continuous evolution must coexist with exact algebra.

The state space satisfying the three basic facts is determined by [3] to be the BC covariance structure (this section only verifies that it meets the five requirements); the machine of the paradigm is therefore specified by structure. A **BC state-space computer** is a physical system that carries the BC covariance relations in finite truncation within finite experimental boundaries (a restricted realization, §2.1) and can prepare, control, and read out its states. The scale of a machine is bounded by its addressable mode set, occupation truncations, and auxiliary degrees of freedom; modes, excitations, phases, couplings, and conserved charges all correspond to measurable physical quantities, and the physical carrier of computation consists of the states the device can reach and operate on.

A run of a BC state-space computer starts from external conditions and passes through five links in order:

1. Boundary formation: pumps, external fields, temperature, chemical potential, selection rules, and couplings to other systems form a boundary through transduction; the boundary selects admissible states, available couplings, dynamics, and observation relations, and the physical form of a program is precisely the constraint-imposing boundary condition;
2. Preparation: available states are placed into the admissible sector;
3. Evolution: the system evolves under the boundary constraints until a stopping event or a sampling time;
4. Readout: terminal instruments produce a record and a post-measurement state;
5. Return: return coupling delivers the record and the post-measurement state into the environment or into the next run.

One and the same machine forms different workspaces under different boundaries, such as arithmetic, probabilistic inference, or a perception–action loop.

2 The State Space and Its Runs

The objects of a run are given in two parts: the machine supplies the reachable states and their intrinsic covariance relations, and the boundary selects one actual run out of these states and relations. This section first defines the finite state space from the global BC carrier, then gives the complete chain of a run.

2.1 The Finite State Space

The intrinsic relations realized by a BC state-space computer begin with the independent composition of distinguishable excitations. Unique factorization of the positive integers

$$n = \prod_p p^{v_p(n)}$$

organizes the global Hilbert carrier as a restricted tensor product along prime directions:

$$\mathcal{H}_{\text{BC}} = \ell^2(\mathbb{N}^\times) \simeq \bigotimes_p' \ell^2(\mathbb{Z}_{\geq 0}). \quad (1)$$

The right-hand side is referenced to the vacuum of each mode and admits only finitely many nonzero occupations. The occupation-number operator of the p -th factor is written N_p ; the basis vector $|n\rangle$ corresponds to the occupation sequence $(v_p(n))_p$. The energy of the system is

$$H = \sum_p (\log p) N_p, \quad H|n\rangle = (\log n)|n\rangle. \quad (2)$$

Independent composition thus corresponds to addition of energy, and time evolution is given by $\sigma_t(A) = e^{itH} A e^{-itH}$.

The system further realizes phase elements $e(r)$ ($r \in \mathbb{Q}/\mathbb{Z}$) and multiplicative isometries μ_n . They satisfy the defining relations of the Bost–Connes algebra [2]:

$$e(r)e(s) = e(r+s), \quad \mu_m \mu_n = \mu_{mn}, \quad \mu_n^* \mu_n = \text{I}, \quad (3)$$

$$\mu_n e(r) \mu_n^* = \frac{1}{n} \sum_{ns=r} e(s), \quad \sigma_t(e(r)) = e(r), \quad \sigma_t(\mu_n) = n^{it} \mu_n. \quad (4)$$

Together these relations give the BC covariance structure of energy, phase, multiplication, and time evolution.

A concrete device realizes only a finite mode set Λ , finite occupation truncations, and finitely many couplings. Write $\mathcal{H}_\Lambda$ for the Hilbert carrier spanned by these addressable degrees of freedom, and $\mathcal{H}_{\text{aux}}$ for the auxiliary degrees of freedom that carry dilations, measurement, or coarse-graining. If, after projection to $\mathcal{H}_\Lambda$, some finite operation acts many-to-one, the device must first realize an isometric or unitary dilation on $\mathcal{H}_\Lambda \otimes \mathcal{H}_{\text{aux}}$ and then read out, discard, or reset the auxiliary degrees of freedom.

Finite modes may also take the local fiber of a general prime p at truncation depth K ,

$$\mathcal{H}_{p,K} = L^2(\mathbb{Z}/p^K \mathbb{Z}, \mathbb{C}),$$

whose translations, phases, involution, and Fourier/Weil actions are compatible with the covariance relations above (results at the Weil and Hecke layers take odd primes; $p = 2$ requires a metaplectic extension, not used in this paper). $\mathcal{H}_{\text{BC}}$ gives the global composition rules; $\mathcal{H}_\Lambda$ and the local fibers give the spaces in which a finite device actually carries these relations.

The Hilbert carrier fixes the space in which states can be represented; the device's preparation, control, and readout capabilities further restrict which states can actually be reached. Write

$$\mathfrak{D}_\Lambda = \{\rho \geq 0 : \text{Tr } \rho = 1, \rho \text{ preparable, controllable, and readable by the finite device}\}. \quad (5)$$

The support of a state ρ is contained in $\mathcal{H}_\Lambda$; while auxiliary degrees of freedom have not yet been read out or discarded, ρ is temporarily represented by an extended state on $\mathcal{H}_\Lambda \otimes \mathcal{H}_{\text{aux}}$, and elements of $\mathfrak{D}_\Lambda$ are its reductions to $\mathcal{H}_\Lambda$. What a run actually transports and changes is $\rho \in \mathfrak{D}_\Lambda$; the boundary and the run condition only select which parts of it can be prepared, evolved, and observed.

For a finite mode set Λ and occupation truncations $0 \leq v_p \leq L_p$, states can be described in three mutually compatible coordinate systems.

- (1) **Occupation coordinates.** The basis vectors

$$|\mathbf{v}\rangle = \bigotimes_{p \in \Lambda} |v_p\rangle_p, \quad \mathbf{v} = (v_p)_{p \in \Lambda},$$

record the occupations of the addressable modes. When $\mathbf{v} = \mathbf{v}_\Lambda(n)$,

$$n = \prod_{p \in \Lambda} p^{v_p(n)}, \quad \mathbf{v}_\Lambda(nm) = \mathbf{v}_\Lambda(n) + \mathbf{v}_\Lambda(m).$$

Occupation vectors can also carry symbols, constraint labels, or sensor features directly, without being interpreted in advance as place-value integers.

- (2) **Finite phase coordinates.** For a modulus q , a residue class $a \in \mathbb{Z}/q\mathbb{Z}$ can be represented by the position state $|a\rangle$ or the phase-slope state $F_q|a\rangle$, where

$$F_q|a\rangle = \frac{1}{\sqrt{q}} \sum_{k \in \mathbb{Z}/q\mathbb{Z}} e^{2\pi i a k/q} |k\rangle, \quad V_c F_q|a\rangle = F_q|a+c\rangle, \quad (6)$$

and $V_c|k\rangle = e^{2\pi i c k/q} |k\rangle$. Composition of phase conditions realizes addition mod q .

- (3) **Correlated-state coordinates.** A general state can be written as

$$\rho = \sum_{\mathbf{v}, \mathbf{w}} \rho_{\mathbf{v}, \mathbf{w}} |\mathbf{v}\rangle \langle \mathbf{w}|,$$

with an equivalent expansion in finite phase coordinates. Diagonal terms give the occupation distribution; off-diagonal terms and cross-mode correlations give phase, superposition, and entanglement.

These coordinates describe different observable relations of one and the same physical state; Fourier/Weil actions change coordinates and measurement bases without creating a new kind of state.

2.2 The Complete Chain of a Run

The unit in which the machine takes on tasks is the **run**: the complete physical process that starts from external conditions, passes through boundary formation, preparation, evolution, and readout, and returns to external conditions. At the start of the k -th run the system holds the pre-run available state $\rho_k^- \in \mathfrak{D}_\Lambda$ and the environment holds the external condition $s_{\text{phys}}^{(k)}(t)$; the device forms the boundary, prepares the initial state, completes the constrained evolution and

the terminal readout, and returns the record and the post-measurement state to the environment. The complete chain of a run is

$$\boxed{\begin{array}{c} (s_{\text{phys}}^{(k)}, \rho_k^-) \xrightarrow{\mathcal{T}_\Lambda} (b_k, \mathcal{Q}[b_k], \rho_k^-) \xrightarrow{\mathcal{P}_{b_k}} \rho_{k,0}, \\ \rho_{k,0} \xrightarrow{\Phi_{\tau_{b_k},0}^{b_k}} \rho_{k,*} \xrightarrow{\mathcal{I}_{b_k}} (r_k, \rho_{k,+}) \xrightarrow{\mathcal{W}_{b_k}} (s_{\text{phys}}^{(k+1)}, \rho_{k+1}^-) \end{array}}. \quad (7)$$

The following subsections define, in the order of the arrows, the boundary and the run condition, preparation and evolution, monitoring and stopping, terminal readout, and return coupling.

2.3 Boundary Formation and the Run Condition

Light, sound, force, temperature, chemical potential, or the port state of another finite system among the external conditions can enter a run only through the device's transduction coupling. This coupling turns external conditions into the system's boundary:

$$\mathcal{T}_\Lambda : s_{\text{phys}}^{(k)}(t) \mapsto b_k(t). \quad (8)$$

The boundary $b_k(t)$ is the complete arrangement in time of the boundary's physical parameters, including pumps, external fields, selection rules, the retention or release of conserved charges, and connections to other systems. External variables not coupled by $\mathcal{T}_\Lambda$ do not enter the present run.

The boundary first selects which states the system may enter during the run. Write

$$\mathcal{S}_b = \{\rho \in \mathfrak{D}_\Lambda : \rho \text{ satisfies the support, conservation, and selection conditions of boundary } b\} \quad (9)$$

for the set of admissible states; expectation constraints $\text{Tr}(\rho C_a[b]) = c_a$, spectral-support restrictions, and superselection conditions are its concrete forms.

The same boundary also selects the constrained evolution Φ^b , the set of terminal observables $\mathcal{O}_b$, the in-run monitoring instrument $\mathcal{J}_{\text{stop},b}$, and the stopping criterion h_b . These objects combine into the run condition

$$\mathcal{Q}[b] = (\mathcal{S}_b, \Phi^b, \mathcal{O}_b, \mathcal{J}_{\text{stop},b}, h_b). \quad (10)$$

$\mathcal{Q}[b]$ prescribes the sectors the system may enter, the evolution it follows, and the manner in which it accepts observation; it does not specify the actual initial state, trajectory, or record of the present run.

2.4 Preparation and Constrained Evolution

Once the run condition is fixed, the device places the pre-run state ρ_k^- into the admissible sector by a preparation process:

$$\mathcal{P}_{b_k} : \rho_k^- \mapsto \rho_{k,0} \in \mathcal{S}_{b_k}. \quad (11)$$

The initial state may be formed by cooling or reset, written into occupations and phases by particle or energy exchange, inherited from the previous run's post-measurement state, or transferred in from another finite system. The device's preparation range, resources, and errors bound the actual domain of $\mathcal{P}_{b_k}$.

After preparation the system evolves under the device's BC covariance structure, intrinsic couplings, and boundary conditions:

$$\rho_k(t) = \Phi_{t,0}^{b_k}(\rho_{k,0}), \quad t \geq 0. \quad (12)$$

In a closed system $\Phi_{t,0}^{b_k}$ may be unitary evolution; when auxiliary degrees of freedom or the environment are coarse-grained, it is the corresponding physical channel; when monitoring is present during the run, it denotes the conditional evolution including measurement back-action. The concrete generator is given jointly by the device's materials, geometry, energy exchange, and boundary couplings.

Whether a state relation is retained is decided by this evolution. For the physical evolution $\Phi_{t,0}^b$ under boundary b , if an observable X satisfies on $0 \leq t \leq T$

$$(\Phi_{t,0}^b)^*(X) = X, \quad (13)$$

then its expectation is retained in that time window. If the spectral projections of X also satisfy the same relation, and monitoring and terminal readout are compatible with these projections, its spectral values can be read repeatedly. Conserved charges and decoupled sectors thereby provide the part of the state that can be retained across runs; unprotected occupations, phases, and correlations may still serve as working states (see §3.7 for the memory classification), but are limited by coherence and dissipation timescales.

2.5 Monitoring and Stopping

In-run monitoring decides when evolution passes to terminal readout. The monitoring instrument $\mathcal{J}_{\text{stop},b}$ produces a record process $Y_b(t)$; the event-triggered stopping time is

$$\tau_b = \inf\{t \geq 0 : h_b(Y_b(t)) = 1\}. \quad (14)$$

Monitoring back-action is contained in Φ^b . When the monitoring instrument is compatible with the monitored spectral projections, monitoring constitutes a QND process. Scheduled sampling requires no continuous monitoring and takes τ_b to be a given time.

2.6 Terminal Readout and Statistical Relations

After the stopping event or the sampling time arrives, the pre-readout terminal state is

$$\rho_{k,*} = \Phi_{\tau_{b_k},0}^{b_k}(\rho_{k,0}). \quad (15)$$

The terminal quantum instrument $\{\mathcal{I}_{b,r}\}_r$ reads out this state. A single run produces a record r_k with probability and post-measurement state

$$p_b(r \mid \rho_{k,*}) = \text{Tr } \mathcal{I}_{b,r}(\rho_{k,*}), \quad \rho_{k,+} = \frac{\mathcal{I}_{b,r_k}(\rho_{k,*})}{p_b(r_k \mid \rho_{k,*})}. \quad (16)$$

The record r_k is one actual readout. Repeated experiments under the same run condition form probability laws, expectations, spectra, correlations, responses, and conserved charges; write $\mathcal{R}[b; \rho_{k,*}]$ for these verifiable statistical objects compatible with $\mathcal{O}_b$. Terminal records, post-measurement states, and the statistical objects of repetition are thereby kept distinct. The Born form of measurement probability is used in this paper through the quantum-instrument formula; its axiomatic load is carried by the KMS states and modular structure of the BC system [3].

2.7 Return Coupling and the Run Law

After terminal readout, the device sends the actual record and the post-measurement state back to the environment through return coupling:

$$\mathcal{W}_{b_k} : (r_k, \rho_{k,+}) \rightsquigarrow (s_{\text{phys}}^{(k+1)}, \rho_{k+1}^-). \quad (17)$$

Return coupling may drive external fields or actuators, pass a port state to another finite system, or form legacy displays, files, or audit records. The arrow $\rightsquigarrow$ retains the randomness introduced by measurement and environmental coupling. If the return ends in a display or a record, (7) gives one standalone run; if it changes the environment, the boundary, or the next run's available state, it forms sequential feedback or a perception–action loop.

Mathematically, $\mathcal{T}_\Lambda$ is a measurable map from external conditions to boundaries, $\mathcal{P}_b$ and Φ^b are quantum channels, $\mathcal{I}_b$ is a quantum instrument, and $\mathcal{W}_b$ is a probability kernel from post-measurement state and record to the next run's external conditions. Given a boundary b and a pre-run state ρ^- , the composition of these maps therefore determines the joint probability law of the single-run record, the post-measurement state, and the post-return external condition:

$$\text{Run}_M(b, \rho^-) = \text{Law}(r, \rho^+, s'_{\text{phys}} \mid b, \rho^-). \quad (18)$$

A single run produces one sample of this law; repeated runs estimate its statistical relations. The next section defines task, program, computation, correctness, and resources on top of this run law.

3 Computational Semantics

A run explains how the machine produces records, post-measurement states, and environmental responses under a stretch of boundary. Computation must further specify the task the run serves, the reusable part of the boundary, the way results are interpreted, and the conditions of completion and acceptance. These objects apply to any workspace carried by a BC state-space computer and presuppose neither arithmetic inputs nor thermalized runs.

3.1 Task, Run, and Computation

A task first specifies the verifiable physical relations the machine is to realize. Write

$$\mathcal{T} = (\mathbf{X}, \mathbf{Y}, \Gamma_{\mathcal{T}}, \varepsilon), \quad (19)$$

where $\mathbf{X}$ is the class of admissible input conditions or states, $\mathbf{Y}$ the class of reportable results, $\Gamma_{\mathcal{T}}(x)$ the target statistical relations and the state or environment postconditions required for input $x \in \mathbf{X}$, and ε the acceptance tolerance. $\Gamma_{\mathcal{T}}$ may specify target probability laws, expectations, spectra, responses, or conserved charges; it may also specify the acceptance class of single records, or the sets that post-measurement states or the closed-loop environment must enter.

Equation (18) characterizes the probability law of a single run: the system evolves under a stretch of actual boundary and produces one sample of that law. A **computation** fixes a task $\mathcal{T}$, a boundary program π (§3.2), and an input x , and organizes one or more runs until the completion rule yields a report or a failure. Computation does not require different runs to

follow the same microscopic trajectory, only that their reportable objects satisfy the same task specification.

Given a machine M , the operational semantics of a program π is defined as

$$\boxed{\llbracket \pi \rrbracket_M(x) = \text{Law}(r, \rho^+, s'_{\text{phys}} \mid \pi, x)}. \quad (20)$$

The single record r is a sample of this law, the post-measurement state ρ^+ retains the system's post-run condition, and s'_{phys} records the environmental condition after return coupling. The environment and control parameters of the boundary decomposition (21) are suppressed in the formula: the full conditioning set is (π, x, e, u, ρ^-) ; in this paper e, u are fixed to a given class of environmental conditions and family of operating schedules, and ρ^- to the input class of the preparation protocol, with correctness and completion criteria stated uniformly over that class. When probabilities, expectations, spectra, or responses are needed, the computation additionally includes repetition counts and estimation rules; that the machine finishes a single run and that the task reaches statistical precision are therefore two different statements.

3.2 Boundary Program, Input, Environment, and Control

The actual boundary is shaped at once by task choice, input, environment, and operating schedule. So that one program can handle different inputs, decompose the boundary-formation relation as

$$b_{\pi, x, e, u}(t) = \mathcal{B}_M(\pi, x, e(t), u(t)). \quad (21)$$

Here π is the **boundary program**, specifying the reusable admissible couplings, constraint families, preparation protocols, monitoring and readout protocols, result interpretation, and return interfaces; x is the physical condition or input state to be processed in the present run; $e(t)$ is the environmental condition that enters the device boundary without being specified by the task; $u(t)$ is the operating control, comprising pumps, cooling, external fields, coupling switches, and sampling schedules; and $\mathcal{B}_M$ is the concrete device's boundary-realization relation. The four together form the physically occurring $b(t)$.

A program is defined by reusable relations, not by the actual records of any particular run. The same π with different x forms different boundaries; the same input can also run under different environments and control schedules. An environmental variable not coupled by the device's $\mathcal{T}_\Lambda$ or $\mathcal{B}_M$ does not enter the computation. The control $u(t)$ may branch conditionally on earlier QND records, but it may not specify the system's trajectory transition by transition in advance from the target answer.

The input x presupposes no symbolic form. One finite device can accept inputs of four different physical origins: an already-given occupation vector, phase/residue coordinates mod q , a port state passed from another finite system, and external features transduced by sensors; they enter the occupation, finite phase, or general correlated-state coordinates of §2.1. Once the input has entered the run through the boundary-formation relation, the boundary-induced $\mathcal{Q}[b]$ and $\mathcal{P}_b$ constitute the configuration of the present run.

The control $u(t)$ constitutes the run's control surface: the temperature schedule $\beta(t)$ provides thermalization timescales, switching sequences of couplings impose piecewise constraints, and post-readout pump switching provides conditional branching; the timetable is set by the relaxation spectrum and the target precision, and time-dependent switching of selection rules realizes the piecewise imposition of constraints. This mode of control belongs to the measurement-driven and adiabatic class [8].

Each item of the boundary program already has its physical landing point in §2: admissible couplings and constraint families enter $\mathcal{Q}[b]$ and select Φ^b , preparation protocols enter $\mathcal{P}_b$, monitoring and readout protocols enter $\mathcal{J}_{\text{stop},b}$, h_b , and $\mathcal{I}_{b,r}$, return interfaces enter $\mathcal{W}_b$, and result interpretation acts at the record layer (§3.4). π is the joint declaration of these reusable relations; $b(t)$ is its physical instance under given input, environment, and control.

3.3 Physical Compilation

A task specification must be turned into a boundary program the device can actually realize. For a machine M , **physical compilation** is the partial map

$$\mathfrak{C}_M : \mathcal{T} \longmapsto \pi. \quad (22)$$

It is defined only when the states, couplings, monitoring, and readout the task requires all fall within the machine’s realizable range. Compilation accomplishes, in order:

- (1) mapping the input conditions in $\mathbf{X}$ to transducible boundary variables or preparable states;
- (2) mapping the target relations in $\Gamma_{\mathcal{T}}$ to admissible states, constraints, and realizable couplings;
- (3) mapping the result class $\mathbf{Y}$ to terminal observables, quantum instruments, and interpretation maps;
- (4) mapping the acceptance tolerance to monitoring, sampling, repetition counts, and completion rules;
- (5) rejecting the compilation when modes, gaps, couplings, coherence times, or readout resolution are insufficient.

The instruction sequences produced by classical compilers are therefore not basic objects here. What $\mathfrak{C}_M$ produces are the conditions that allow a physical process to happen, together with its observation interface; the actual intermediate trajectory is produced by Φ^b . If an external controller sets the target transition rates one by one, or writes the answer directly into the target sector, the resulting process is **protocol emulation** (our name for the practice of writing in the target dynamics rate by rate, or the target result directly); it tests only the protocol itself and does not realize the physical program declared by (22).

3.4 Results, Outputs, and Correctness

Four different objects can remain when a computation ends: the single record r_k , the post-measurement state $\rho_{k,+}$, the statistical objects formed by repeated runs, and the physical response after return coupling. The program’s interpretation map

$$\mathcal{D}_\pi : (r, \rho^+, s'_{\text{phys}}) \longmapsto y \in \mathbf{Y} \quad (23)$$

extracts only what the task specification allows to be reported; its pushforward law is written $(\mathcal{D}_\pi)_* \llbracket \pi \rrbracket_M(x)$.

Records themselves are formed by three kinds of physical readout: response-function readout outputs values of analytic functions by amplitude measurement with a coherent probe,

sector-occupation readout reports the occupation of a target sector, and Born-statistics readout estimates $|\cdot|^2$ by repeated sampling. Output precision is tunable along the two continuous parameters of temperature and sample size, with no discrete jumps in word length; the corresponding sample costs and readout-confusion constraints are accounted for in §3.9 and §3.10, respectively.

The task distance (an operational distance) $d_{\mathcal{T}}$ is chosen by result type: probability outputs may take total variation or Wasserstein distance, expectation and spectrum outputs take the corresponding error norms, state outputs take trace distance or infidelity of the expected post-measurement state (per-sample criteria are stated through acceptance sets), and closed-loop control takes the distance of the environmental variable to the target set. A program is correct on input x if and only if

$$\boxed{d_{\mathcal{T}}((\mathcal{D}_{\pi})_* \llbracket \pi \rrbracket_M(x), \Gamma_{\mathcal{T}}(x)) \leq \varepsilon}. \quad (24)$$

If the specification further contains a state acceptance set $\mathcal{S}_{\text{acc}}(x)$ or an environmental target set $\mathcal{E}_{\text{acc}}(x)$, the post-measurement state and the post-return external condition are additionally required to fall into those sets with prescribed probability.

Single-run correctness means that some r_k falls into the record acceptance class given by the task specification $\Gamma_{\mathcal{T}}$; statistical correctness means that the distribution or estimators of repeated runs satisfy (24); physical correctness further requires the post-measurement state or the actuator response to satisfy the postconditions. The three can occur together, but a favorable single record cannot substitute for statistical correctness, nor internal statistical correctness for the physical response at the actuation end.

3.5 Completion, Stopping, and Failure

The stopping time τ_b is a temporal object internal to a single run; completion of a computation is a criterion on reportable results. Completion rules fall into four classes:

- (1) **Event completion.** The monitoring record first satisfies $h_b(Y_b(t)) = 1$ and the run passes to terminal readout; protocols with post-readout record verification take verification success as the completion event (Las Vegas type).
- (2) **Stabilization completion.** The target observable varies less than a threshold within a given time window and satisfies the corresponding state or response conditions.
- (3) **Statistical completion.** Repetition brings estimation errors, confidence intervals, or hypothesis tests to the precision the task prescribes.
- (4) **Closed-loop completion.** Return coupling and actuators have brought the environmental state or the control error into the target set.

Stopping criteria are implemented by the control surface as time schedules: scheduled readout corresponds to a fixed τ_b ; when an event criterion is first met, the control surface closes or switches the relevant pumps, pinnings, and couplings and hands the system to terminal readout. Fixed-time sampling only fixes the readout time of a single run and does not by itself constitute task completion; continuous sensing and servo loads report persistently over rolling windows and are not required to enter a terminal state.

When a run reaches its resource ceiling without meeting the completion rule, the result is timeout or non-completion. Preparation failing to enter the admissible sector, coherence lost

before readout, monitoring misjudgment, device-acceptance failure (§8.4), and return-actuation failure are distinct failure events. A finite low-temperature run that misses the zero-penalty sector can only report its current samples and energy floor; by itself it cannot decide that the constraints are unsatisfiable. Low-temperature concentration and metastable retention in finite systems are likewise not a non-analytic phase transition.

3.6 Dynamical Modes of Running

The constrained evolution Φ^b is the total dynamics of a run and is not defined by thermalization alone. A boundary program can select four basic modes.

- (1) **Coherent mode.** A closed or nearly closed system preserves phase, superposition, and entanglement under unitary evolution; results form through terminal interference or response readout.
- (2) **Relaxation mode.** The system exchanges energy with the environment and approaches stable states, Gibbs states, or low-temperature concentrated sectors; completion time is governed by spectral gaps, energy barriers, and mixing properties.
- (3) **Monitored mode.** Monitoring records and measurement back-action enter conditional evolution; the trajectory depends on $Y_b[0, t]$, and stopping can be event-triggered.
- (4) **Feedback mode.** The record or post-measurement state of run k changes b_{k+1} ; multi-segment dynamics is closed by physical return coupling.

One stretch of running may combine these modes in segments: for example, a coherent probe first forms a response, QND monitoring then selects a conditional branch, and a relaxation segment stabilizes the output. Beyond the dynamical modes, tasks divide by how they use the machine into three **load modes**: evaluation outputs values of analytic functions by response measurement on probe states, solving outputs solutions of constraints by concentration of weight toward the lowest-penalty sector, and search outputs satisfying items by configuration updates with conditional verification. Load modes prescribe the form of task and readout; dynamical modes prescribe how Φ^b runs: evaluation has the coherent mode as its main component, solving the relaxation mode, and search is organized by the monitored and feedback modes. The concrete forms of the three loads appear in §5.1.2–§5.1.4.

3.7 Native Memory

The information a machine can use across time is determined by the retention relations of physical states. The state ρ carries the data; the boundary's $\mathcal{Q}[b]$ prescribes in which coordinates data can appear, be retained, and be read out. Data may be prepared by the current boundary, inherited from the previous run's post-measurement state, or transferred in from another finite system; its occupations, phases, correlations, and conserved charges all belong to ρ . Native memory falls into four classes:

- (1) **Working states.** Unprotected occupations, phases, and correlations participate in the current run within finite coherence or dissipation times.
- (2) **Protected states.** Conserved charges, decoupled sectors, or spectral projections satisfying (13) are retained within the given boundary and time window.

- (3) **Stable states.** Dynamical attraction, penalty gaps, or energy barriers keep the state in a stable or metastable sector within a specified error and time window.
- (4) **Environmental states.** The record has been written into another finite system, an external field, an actuator, a legacy display, or an audit medium.

Retention is the evolution itself keeping some observable relation invariant; **saving** is actively writing a result into a carrier recoverable in later runs. Protected states can be retained without continual writing; environmental states usually pass through a saving process. Finite systems have no zero-cost checkpoint/restore of arbitrary states; recoverable content, retention time, writing error, and reset cost are all given by the concrete carrier.

3.8 Composability

Several computations compose into a larger one through physical interfaces; composition comes in three kinds.

- (1) Serial composition feeds the first program's post-measurement state, record, or environmental response, through the return coupling of its boundary instance, into the second program's boundary-formation relation, written $\pi_2 \circ \pi_1$.
- (2) Parallel composition runs two programs on a joint carrier: if the joint initial state factorizes and boundaries, environments, and conserved charges are unshared, the joint semantics is the tensor product; if the initial state is correlated or anything is shared, joint dynamics and a joint stopping rule must be used.
- (3) Feedback composition sends outputs back through return coupling into an upstream program or a common environment, forming a closed loop.

The physical mechanism of serial and feedback composition is closed by three components: retention-compatible QND readout (post-measurement state unchanged), registers made of decoupled conserved charges (coupling on to write, coupling off to retain, occupation measurement to read), and readout-triggered boundary reconfiguration (conditional branching). The result of an earlier segment enters the later segment's boundary-formation relation through the register; the feedback basis is storage-compatible, and the loop does not disturb retained states.

Independent composition and physically coupled composition have different conditions: independent composition requires the joint initial state, boundary, and evolution all to factorize; coupled composition may exchange energy, occupations, phases, or conserved charges, and may generate classical correlations or entanglement. Two interfaces can dock only when the following hold simultaneously.

- (1) The output state belongs to the downstream reachable input class;
- (2) conserved-charge flows are compatible at the junction;
- (3) the admissible coupling can be realized;
- (4) timing and stopping conditions can be synchronized;
- (5) the downstream side is entitled to read the relevant observables of the upstream output.

Parallelism does not satisfy these conditions automatically just because it is written symbolically as a tensor product.

3.9 Resources and Complexity

The cost of computation is decided by the complete physical chain. For program π , input x , and target error ε , write the resource vector

$$\mathbf{C}_M(\pi, x, \varepsilon) = (T_{\text{trans}}, T_{\text{prep}}, T_{\text{dyn}}, T_{\text{read}}, T_{\text{fb}}; N_{\text{mode}}, D_{\text{aux}}; E_{\text{drive}}, W_{\text{reset}}; \delta_b, \delta_\phi, \delta_{\text{read}}; M, P_{\text{fail}}). \quad (25)$$

The first five entries are transduction, preparation, dynamics, readout, and feedback times; N_{mode} and D_{aux} count addressable modes and auxiliary space; E_{drive} and W_{reset} meter drive energy and reset free energy; the three δ 's meter boundary, phase, and readout precision; M is the repetition count and P_{fail} the failure probability. Relaxation loads must additionally report spectral-gap, mixing, or barrier timescales; coherent loads must report coherence times; the imperative mode (§5.3.2) must count controller, head, addressing, and communication resources.

Complexity belongs to uniform machine families $\{M_n\}$ of growing input size; a single finite device carries no asymptotic statement. Given a task family and a uniform compiler, polynomial-time, sample-complexity, or space-complexity conclusions can be made only when the designated components of (25) satisfy the declared asymptotic bounds in the input length. The size of the state-space dimension is no evidence of computational efficiency; exponentially small gaps, exponential control precision, and exponential sample counts are all computational costs.

3.10 Error Model and Reliability

Errors are classified along the physical chain of the run; their sources and acceptance quantities are:

- (1) transduction and boundary errors: the actual $b(t)$ deviates from the history declared by (21);
- (2) preparation errors: ρ_0 fails to enter the target admissible sector or is too far from the target initial state;
- (3) dynamics and model errors: the actual channel deviates from the declared Φ^b ;
- (4) decoherence and leakage: working-state correlations decay, or the state leaves the reachable coding space;
- (5) monitoring and stopping errors: events are missed or falsely reported, or monitoring back-action exceeds the model;
- (6) terminal readout and statistical errors: instrument bias, classification errors, and finite-sample fluctuations;
- (7) return and actuation errors: $\mathcal{W}_b$ fails to form the required port state or environmental response.

The items are estimated respectively by boundary-parameter errors, state distances, channel distances, leakage rates, stopping misjudgment rates, readout confusion matrices, and control deviations. The reliability of serial composition is given by propagation bounds of these errors across the actual interfaces; one may not simply add per-segment error probabilities while ignoring post-measurement states and feedback amplification.

Symmetry-protected readout only constrains the set of directly reportable operators and does not constitute a general error-correcting code; Galois-invariant readout in the arithmetic workspace (§4.1.4) is its instance. Conserved charges, decoupled sectors, physical redundancy, repeated runs, and statistical estimation carry different reliability roles; for general local noise, code distances, thresholds, or error-mitigation protocols must still be supplied separately.

3.11 Expressive Power, Realizability, and Universality

Whether a task can enter the machine involves four mutually independent questions. First, expressive power requires the task to be writable as transducible inputs, reachable state constraints, realizable boundary relations, and measurable results. Second, physical realizability requires $\mathfrak{C}_M$ to be defined on the task and actually realized by the device. Third, resource feasibility requires (25) to fall within the given resource bounds. Fourth, simulability asks whether the input–output distribution can be approximated by classical or quantum models at corresponding resources. Expressive power does not imply the other three.

For a task class $\mathfrak{F}$, if there exist a uniform machine family $\{M_n\}$ and a uniform compiler such that every $\mathcal{T} \in \mathfrak{F}$ compiles within the prescribed error and resource bounds and satisfies (24), the family is universal relative to $\mathfrak{F}$. The DPRM theorem [6] can establish the expressive power of a class of Diophantine constraint languages; finite-tape Turing simulation can establish the compatibility expressiveness of the imperative mode; neither automatically yields acceptable gap, preparation, addressing, or readout costs. Classical simulability must likewise be judged per task family and per resource quantifier.

In this paradigm a program is a boundary condition rather than data inside the machine; there is no stored-program principle, nor a classical-style universal machine that takes its own program as input; representability between machines is carried by the relative universality above: take the task class to be the family of run laws of the other machine.

Self-reference is thereby physicalized, with no need for prohibition: a description of a program can be written into a state, but making it effective must pass through a measurement channel (an in-run QND conditional branch, or cross-run closure by return coupling; §3.2) and be billed for resources; evolution under a fixed boundary can neither choose nor rewrite its own boundary. Syntactic self-reference and its attendant structures (self-modification, program injection) do not enter the native layer; their roles are taken over in three places: recursion and universality by the feedback mode and the compatibility layer (the imperative mode’s instruction table and compilation, §5.3.2), fixed points by stable states and closed-loop steady states, and diagonalization by the countermodels of the certificate system (Appendix C.1). Limitative theorems do not thereby vanish: when Turing simulation is carried through the compatibility layer (§5.3.2), halting-type undecidability again constrains the task classes of the corresponding machine families.

3.12 The Paradigm Table

The definitions above ground the interface vocabulary of computing in physical objects along the run chain. The second column of the table gives the native semantics, the third the corresponding acceptance objects; these correspondences are an interface index and do not treat the two vocabularies as synonyms.

Interface term	Native object in the BC state-space computer	Acceptance object
Task	$\mathcal{T} = (X, Y, \Gamma_{\mathcal{T}}, \varepsilon)$	Target relations and postconditions
Program	Reusable boundary program π	Realizable couplings, constraints, and observation interfaces
Input	This run's physical condition or input state x	Transduction range and preparation error
Data	Occupations, phases, correlations, and charges of $\rho \in \mathfrak{D}_{\Lambda}$	State distance, leakage, and readability
Run	Preparation, Φ^b , monitoring, readout, and return under b	Joint run probability law (18)
Completion	Event, stabilization, statistical, or closed-loop criterion	Stopping misjudgment, stability window, confidence interval, or control error
Output	Records, post-measurement states, statistical objects, and physical responses	Interpretation map and task distance
Memory	Working, protected, stable, and environmental states	Retention time, write/recovery error, and reset cost
Composition	Serial, parallel, feedback interfaces (independent/coupled dichotomy accounted separately)	Charge conservation, timing, state and readout-entitlement compatibility
Correctness	Operational probability law satisfies the task specification	Eq. (24)
Complexity	Resource vector of physical time, modes, energy, precision, and samples	Asymptotic bounds on (25)
Error	Seven deviation classes along the run chain	State, channel, readout, and control error budgets
Universality	Uniform machine family and compiler relative to a task class	Correctness and uniform resource bounds

The scope of the definitions above is delimited by one thesis. By the argument of [3], the three facts that excitations are countable, energy is additive, and independent composition multiplies uniquely determine $(\mathbb{N}, +, \times)$ and the BC covariance structure upon it, and, under channel identification, give the observed numbers of the Standard Model; the machine definition of this paper takes the first two layers as premises. The thesis states: for every physical system satisfying the three facts, the computational processes it carries are all representable as runs of some BC state-space computer under a corresponding boundary program. The statement is a thesis, not a theorem; it delimits the domain of the paradigm. In which systems the three facts hold is decided by experiment; the decisive predictions of [3] provide the test at the level of fundamental physics.

4 Arithmetic and Interaction Modules

The three preceding sections give, in turn, the definition of the machine, the complete chain of a run, and the computational semantics: §1 fixes the physical form of the machine and its runs, §2 writes a run as the chain $(s_{\text{phys}}, \rho^-) \rightarrow \cdots \rightarrow (s'_{\text{phys}}, \rho'^-)$ and gives the run law, and §3 defines task, program, compilation, and correctness on top of the run law. Up to this point every

arrow of the chain is still a mathematical kind of object: measurable maps, quantum channels, instruments, probability kernels; how the machine actually runs depends on the physical content filled into these arrows.

This section supplies that physical content. The machine's parts belong to two modules: the **arithmetic module** carries the representation and processing of content, and the **interaction module** carries the machine's interaction with the world. Modules do not partition the chain into segments: every run traverses the whole chain, and the modules are the two families of parts the boundary can call upon; workspaces are given by the mix of calls, with the arithmetic workspace dominated by the arithmetic module, the perception–action loop by the interaction module, and probabilistic inference using both.

In a BC state-space computer, arithmetic is the organizing principle of the state space: basis vectors are the positive integers, and occupation vectors are unique factorizations ((1)); independent composition is multiplication, and additive energy is the logarithm ((2)); phase residues carry modular arithmetic, and addition adds no primitive, being assembled from scalings and translations.

The arithmetic module's family of boundary programs exposes this structure as computation: admissible couplings are restricted to the covariant operations μ_n , U_χ , and V, F assemblies, selection rules preserve the tensor structure along prime directions and the conserved charges, thermal contact realizes KMS-locked energy exchange, and readout takes occupations, phases, and responses. The tasks, programs, compilation, and completion criteria of §3 take their values on this family.

The content of the interaction module is the machine's interaction with the world; the external conditions are themselves the object of the task. Its boundary shares are dominated by transduction and return coupling: transduction maps distinguishable features of external fields into boundary, occupation, phase, and constraint increments (evidence energy, §4.2); actuation ports turn records, through admissible feedback relations, into new boundary conditions and drive actuators; monitoring runs persistently over sampling windows. The corresponding task classes are detection and servo stabilization, while inference and sustained tracking use both modules (§5.2–§5.3); completion takes event, stabilization, or closed-loop criteria as the task requires (§3.5), and the correctness of closed-loop tasks is accepted by the distance of the environmental variable to the target set (§3.4).

The arithmetic module consists of four links, ordered along a run:

$$x \longmapsto \mathbf{v}_\Lambda, F_q|a\rangle \xrightarrow{\mu_n, U_\chi, V, F, e^{iHt}} \rho \xrightarrow{\mathcal{L}_\beta} \rho_* \xrightarrow{\tau_b, \mathcal{I}_b} r. \quad (26)$$

The writing and retention of data, the covariant operations and their algebraic closure, relaxation and equilibrium by thermal contact, and scheduled stopping with readout take the four links in turn (§4.1.1–§4.1.4); the interaction module's parts are transduction, evidence, and actuation, distributed along the loop (44), with inter-machine coupling and legacy interfaces as their port forms (§4.2). The instances unfold uniformly by complexity in the next section; that unfolding gives the part form and the instance form of each interface term of the table in §3.12.

4.1 The Arithmetic Module

The parts of the arithmetic module unfold along the four links of (26).

4.1.1 Writing and Storing Data

The data parts take the first link of the run. The arithmetic module's input takes the occupation or phase form of the x defined in §3.2. If the prime support of an integer n is contained in Λ , and its occupation vector is already supplied externally or prepared by the boundary, writing is

$$n \mapsto \mathbf{v}_\Lambda(n) = (v_p(n))_{p \in \Lambda}.$$

If the task gives a in the phase sector mod q , one prepares $F_q|a\rangle$, or equivalently pins its phase slope. Conversely, when only the place-value numeral n is given, obtaining the occupation vector still requires the place-value-to-occupation conversion (cost accounted in §6.6); this cost belongs to the input boundary, not to any zero-cost internal action of the machine.

Once written, data are carried in occupation and phase coordinates, with retention organized by the memory classes of §3.7. In the occupation representation a rational a/b can be represented by two sets of nonnegative occupations, numerator and denominator; per-channel reduction can be organized by the feedback of §5.3.2. Signed occupations $v_p \in \mathbb{Z}$ are the isometric extension from $\mathbb{N}^\times$ to $\mathbb{Q}_+^\times$ and must carry corresponding inverses or auxiliary records in a finite Fock realization. The internal coordinate change between phase and occupation sectors is written F ; in a mod- q cell it is the F_q of (6).

Preparation resources are decided by the factorizability of the coefficients along prime directions. We first fix the normalizability condition, then discuss the cross-mode correlations produced by additive superposition.

Let $c(1) = 1$ and write

$$s_p = \sum_{k \geq 1} |c(p^k)|^2.$$

If c is multiplicative, then

$$|\psi_c\rangle = \sum_n c(n)|n\rangle = \bigotimes_p \left(|0\rangle_p + \sum_{k \geq 1} c(p^k)|k\rangle_p \right), \quad \|\psi_c\|^2 = \prod_p (1 + s_p). \quad (27)$$

The right-hand side converges in the vacuum-referenced incomplete tensor product if and only if $\sum_p s_p < \infty$; this condition is also equivalent to $|\psi_c\rangle \in \ell^2(\mathbb{N}^\times)$. Conversely, if $c(1) = 1$ and $|\psi_c\rangle$ is a product state with all vacuum components nonzero, local factors can normalize the vacuum coefficients to 1, whence $c(n) = \prod_p c(p^{v_p(n)})$, i.e., c is multiplicative. If moreover $c(p^{j+k}) = c(p^j)c(p^k)$, then $c(p^k) = c(p)^k$, giving complete multiplicativity. Hence

$$\boxed{\begin{aligned} c(1) = 1 : \quad \ell^2 \text{ product state} &\iff c \text{ multiplicative and } \sum_p \sum_{k \geq 1} |c(p^k)|^2 < \infty, \\ c \text{ completely multiplicative} &\iff c \text{ multiplicative and } c(p^k) = c(p)^k. \end{aligned}} \quad (28)$$

In the completely multiplicative case the normalizability condition becomes $|c(p)| < 1$ for every p together with $\sum_p |c(p)|^2 < \infty$, and

$$\|\psi_c\|^2 = \prod_p (1 - |c(p)|^2)^{-1}.$$

Product states with $c(1) = 0$ fall outside this equivalence; for example, $|2\rangle$ is a product state whose coefficients are not multiplicative. In general such states arise from finite occupation

translations acting on product states with nonzero vacuum components. In particular, the Gibbs coherent state $\sum_n n^{-\sigma} |n\rangle$ is normalizable for $\sigma > 1/2$ with squared norm $\zeta(2\sigma)$ and can be prepared mode by mode. The probe-preparation cost of the evaluation instance is accounted linearly on this basis (§5.1.2).

The entanglement of additive superposition is determined by the Schmidt criterion. Take

$$|\phi\rangle = c_1|n\rangle + c_2|m\rangle, \quad n \neq m, \quad |c_1|^2 + |c_2|^2 = 1,$$

and split the prime modes into $A \sqcup B$. Write $|n\rangle = |n_A\rangle|n_B\rangle$, $|m\rangle = |m_A\rangle|m_B\rangle$. If $n_A = m_A$ or $n_B = m_B$, the superposition remains a product state and the entanglement entropy across the cut $A|B$ vanishes; if the branches differ on both sides, the two terms already form a Schmidt decomposition, so

$$E_A(\phi) = -|c_1|^2 \log |c_1|^2 - |c_2|^2 \log |c_2|^2. \quad (29)$$

An equal-amplitude two-branch superposition therefore carries $\log 2$ of entanglement only when both sides distinguish the branches. For a single-prime cut $A = \{r\}$ the condition is $v_r(n) \neq v_r(m)$ and $n/m \notin r^{\mathbb{Z}}$. For example, $(|6\rangle + |3\rangle)/\sqrt{2}$ superposes only within the 2-mode and is thus a product state. Whether additive superposition generates cross-mode entanglement depends on whether the two branches differ simultaneously on both sides of the given cut; it cannot be judged from the occupation change of a single mode.

The interface between additive phase vectors and multiplicative characters is given by Gauss sums. In $\mathbb{C}[\mathbb{Z}/q\mathbb{Z}]$ (q prime), take $e_a(n) = e^{2\pi i a n/q}/\sqrt{q}$ and the nontrivial multiplicative character vectors $v_\chi(n) = \chi(n)/\sqrt{q-1}$. Then

$$\langle e_a, v_\chi \rangle = \frac{\bar{\chi}(-a)\tau(\chi)}{\sqrt{q(q-1)}}, \quad |\langle e_a, v_\chi \rangle| = \frac{1}{\sqrt{q-1}} \quad (a \neq 0), \quad (30)$$

while the inner product vanishes for $a = 0$. There are $q-2$ nontrivial characters, spanning $\{f : \text{supp } f \subset (\mathbb{Z}/q\mathbb{Z})^\times, \sum f = 0\}$ of dimension $q-2$; the two families are therefore not a pair of mutually unbiased bases of $\mathbb{C}^q$. What (30) establishes exactly is the flat modulus interface on the generic block, with amplitude $(q-1)^{-1/2} = q^{-1/2}(1 + O(q^{-1}))$. The strictly mutually unbiased pair in this space is the position basis and the Fourier basis.

Storage is a boundary-protected relational structure. If a charge does not participate in the current constrained evolution, it retains its written value within the corresponding time window. After decoupling a character sector from the pump network, the occupation of that sector becomes a conserved charge; writing is done by briefly opening the coupling, retention by keeping it closed, and readout is occupation-number measurement. By

$$[N_p, H] = 0, \quad [N_p, U_\chi] = 0 \quad (31)$$

and the absence of that sector's $\mu^{(*)}$ terms from the decoupled generator, N_p is a constant of motion; measurement of its spectral projections is compatible with the subsequent decoupled evolution, giving QND readout [25].

Retention time under residual coupling is computed only in a specified localized noise model. Let $\beta > 1$, and take a translation-invariant, finite, symmetrized step measure ν_s on $G = (\mathbb{Z}/q^K\mathbb{Z})^\times$. For character observables a_k ,

$$\mathcal{L}_\beta(a_k) = -\text{Re } D_k(\beta) a_k, \quad \text{Re } D_k(\beta) = \sum_{(n,q)=1} \Lambda(n) (1 - \text{Re } \chi_k(n)) n^{-\beta}. \quad (32)$$

For nontrivial χ_k the Dirichlet distribution of primes makes the sum strictly positive, and expectations decay as $e^{-t \operatorname{Re} D_k}$. The equivalent closed form is

$$\operatorname{Re} D_k(\beta) = -\frac{\zeta'}{\zeta}(\beta) - \frac{\log q}{q^\beta - 1} + \operatorname{Re} \frac{L'}{L}(\beta, \bar{\chi}_k) > 0. \quad (33)$$

This conclusion does not apply to the non-unitary isometries at the raw Fock layer; residue-class localization also fails to preserve the modular flow given by $H = \log n$. If both the time scale and the dissipation are needed, one must keep the joint tree model of residue classes and valuation/Busemann coordinates. Equation (32) is therefore a retention-time specification within the localized coupling window, not a universal formula for arbitrary KMS baths; the retention of sequential-program registers within the residual-coupling window is priced by it (§5.3.2).

Passive storage in finite systems is provided by conserved charges or energy barriers. With finite modes and finite truncation the partition function is analytic in β and no genuine non-analytic phase transition occurs; at low temperature the retention time of metastable sectors is controlled by barriers, typically with Arrhenius scaling. Logical erasure is compressing an unknown logical distribution into a standard state. For iso-energetic degenerate memory the heat absorbed by the environment satisfies

$$Q_{\text{env}} \geq k_B T H(p), \quad (34)$$

a uniform bit giving $k_B T \log 2$; the minimal work of a general nondegenerate process satisfies $W \geq F(\rho_{\text{out}}) - F(\rho_{\text{in}})$. Heating a low-entropy state into a high-entropy one is randomization or reset preparation, not logical erasure.

4.1.2 The Operation Algebra

The written data are transformed by operations. The arithmetic module's operations come from the finite restriction of (3)–(4): multiplicative isometries change occupations, character phases probe multiplicative relations, and finite Fourier/Weil actions change phase and position coordinates. The diagonal action of character probing is written $U_\chi |n\rangle = \chi(n) |n\rangle$ (χ a Dirichlet character mod q , with $\chi(n) = 0$ when $q \mid n$, so U_χ is a contraction at the Fock layer and unitary on the coprime sector). We now give the algebraic closure of this operation set.

The pivot of the closure is one crossed relation (full closure up to the SL_2 Weil representation appears at the end of Appendix B). Verified basis vector by basis vector on $|n\rangle$:

$$U_\chi \mu_q |n\rangle = \chi(qn) |qn\rangle = \chi(q) \chi(n) |qn\rangle = \chi(q) \mu_q U_\chi |n\rangle, \quad \text{i.e.} \quad U_\chi \mu_q = \chi(q) \mu_q U_\chi. \quad (35)$$

Step 1, the operation group. After localization mod q , the additive translation $S : n \mapsto n + 1$ (assembled from the operation set: $S = F^{-1} V_1 F$, verified componentwise by $V_1 F |a\rangle = F |a + 1\rangle$) and the multiplicative scaling $M_g : n \mapsto gn$ (the mod- q localization of μ_g ; g a generator of $\mathbb{F}_q^\times$) satisfy

$$M_g S M_g^{-1} : n \mapsto g(g^{-1}n + 1) = n + g, \quad \text{i.e.} \quad M_g S M_g^{-1} = S^g, \quad (36)$$

and generate the affine group $\text{Aff}(\mathbb{F}_q) = \mathbb{F}_q \rtimes \mathbb{F}_q^\times$. Its irreducible representations are $q - 1$ one-dimensional ones (through the multiplicative quotient) and a unique $(q - 1)$ -dimensional one (induced from the additive characters); the machine's working carrier $\mathbb{C}^q$ decomposes as the trivial component plus that $(q - 1)$ -dimensional representation. The two maximal abelian subalgebras, position-diagonal and phase-diagonal, are conjugate to each other on $\mathbb{C}^q$ by F .

Step 2, the interface maps multiplicative characters back into their own family. Computing Fv_χ pointwise,

$$(Fv_\chi)(m) = \frac{1}{\sqrt{q}} \sum_n \chi(n) e^{2\pi i m n / q} = \frac{\tau(\chi)}{\sqrt{q}} \bar{\chi}(m),$$

that is,

$$Fv_\chi = \frac{\tau(\chi)}{\sqrt{q}} v_{\bar{\chi}}, \quad \left| \frac{\tau(\chi)}{\sqrt{q}} \right| = 1. \quad (37)$$

The nontrivial real character (the quadratic-residue symbol, which exists and is unique) satisfies $\bar{\chi} = \chi$ and is a strict eigenvector of F ; its eigenvalue $\tau(\chi)/\sqrt{q}$ equals 1 ($q \equiv 1 \pmod{4}$) or i ($q \equiv 3 \pmod{4}$) by the Gauss sign theorem. Within the family of nontrivial multiplicative characters, the only nontrivial direction with $\bar{\chi} = \chi$ is the quadratic character, so the single character direction preserved by F in that family is given by the quadratic structure.

Step 3, conditional determinacy of the bath. On the local model $L^2((\mathbb{Z}/q^K\mathbb{Z})^\times)$, a translation-invariant Markov generator is determined by its step measure ν , and character diagonalization gives all its eigenvalues $\hat{\nu}(\chi_k)$. If detailed balance, the boundary geometry, and the chosen response functions jointly specify $\hat{\nu}$ on the whole dual group, Fourier inversion

$$\nu(g) = \frac{1}{|G|} \sum_k \hat{\nu}(\chi_k) \overline{\chi_k(g)} \quad (38)$$

recovers ν uniquely. Taking $\nu(g) = \sum_{n \equiv g} \Lambda(n) n^{-\beta}$, the generator is compatible with these conditions; its microscopic origin must still be checked separately against the weak-coupling limit, the branching geometry, and the device data of Appendix B.

Step 4, assembling addition. Arithmetic addition adds no element to the operation set; it is composed from existing actions. Within a mod- q channel a number has two encodings exchanged by F : the position encoding $|a\rangle$ and the phase encoding $F|a\rangle$ (whose n -th component is $e^{2\pi i a n / q}$, the slope being the value). In phase encoding the addition gate is the additive character V_c : componentwise $e^{2\pi i c n / q} \cdot e^{2\pi i a n / q} = e^{2\pi i (a+c)n / q}$, so $V_c F|a\rangle = F|a+c\rangle$, and F^{-1} reads back $(a+c) \pmod{q}$; in position encoding the addition gate is its F -conjugate $S^c = F^{-1} V_c F$. Coprime modulus sets $\{q_1, \dots, q_s\}$ (with $\prod_i q_i$ covering the dynamic range) run channelwise in parallel; the juxtaposition of residue components is precisely the multiplicative side's tensor-product decomposition, a native structure, while reconstruction to place value belongs to output conversion (§8.2). The conjugation (36) yields assembly acceleration: following the binary decomposition of a , S^a composes from alternating scalings and single-step translations in $O(\log a)$ actions; place-value machines assemble multiplication from shifts and additions, this machine assembles addition from scalings and translations, and the cost for each paradigm to simulate the other's operation is the same logarithmic composition of its own native actions. The attribution of costs is clean: for definite inputs the residues are definite values, the channels are independent, and no entanglement is generated; what pays the entanglement bill is addition on superposed states (entanglement computed in the same form as (29)), and the amplitude cost (30) of crossing the flat modulus interface hangs on quantum superposition, not on addition itself. The residue number system, representing integers by coprime residues in parallel, thereby acquires its algebraic locus: it is the shadow of this machine's addition assembly in the place-value world, a patch there, a natural subsystem here.

4.1.3 Operating Environment and Relaxation Spectrum

Operations carry the coherent layer of transformations; the dissipative layer of a run is carried by thermal contact. Bath exchange appears only when the corresponding boundary, geometry, and energy-exchange conditions hold; write $\mathcal{L}_\beta$ for its effective generator. This subsection first obtains the equilibrium target from maximum entropy, then gives the conditional microscopic mechanism and spectral structure of $\mathcal{L}_\beta$.

One maximum-entropy step gives the origin of the partition function. On all density operators, fix the energy expectation $\langle H \rangle = E$ and perform the maximum-entropy variation [32]: the von Neumann entropy is strictly concave on this convex set, the maximizer is unique, and it is the Gibbs state $\rho_\beta = e^{-\beta H}/Z$, with β the Lagrange multiplier. By the modewise additivity of (2), the Gibbs state factorizes into modewise geometric distributions $P_p(k) = (1 - z_p)z_p^k$, whose parameters can be verified mode by mode. The single-mode entropy sums directly:

$$S(z) = -\sum_{k \geq 0} P(k) \log P(k) = -\log(1 - z) - \frac{z}{1 - z} \log z. \quad (39)$$

Vary under the energy constraint $\sum_p (\log p) \frac{z_p}{1 - z_p} = E$. The two derivatives are

$$\frac{dS}{dz} = -\frac{\log z}{(1 - z)^2}, \quad \frac{d}{dz} \frac{z}{1 - z} = \frac{1}{(1 - z)^2}, \quad (40)$$

and the Lagrange condition $dS/dz_p = \beta (\log p) d\langle N_p \rangle/dz_p$ reduces to $-\log z_p = \beta \log p$, i.e.,

$$\boxed{z_p = p^{-\beta}, \quad Z = \prod_p \frac{1}{1 - p^{-\beta}} = \zeta(\beta).} \quad (41)$$

Here $\beta > 1$ is required for a trace-class Gibbs state; for given finite energy E , the function $E = -\zeta'(\beta)/\zeta(\beta)$ is strictly monotone on $(1, \infty)$, so β is unique. What the strict concavity of entropy uniquely determines is the equilibrium state and its statistics, not the relaxation generator. A uniform-rate birth–death chain and a bosonically enhanced chain sharing the same geometric steady state $\pi_k = (1 - z)z^k$ can have spectral gaps $\kappa(1 - \sqrt{z})^2$ and $\kappa(1 - z)$ respectively; the dynamics is determined within the corresponding model class only after the jump graph, edge weights, KMS ratio, and overall timescale are all specified. The target distributions of the solving and inference instances are precisely this Gibbs family (§5.1.3, §5.3.1).

Step 3 of §4.1.2 gives the conditional uniqueness of the bath rate spectrum under equilibrium-family matching (given the complete response on the dual group, inversion is unique); this subsection gives that bath’s *conditional* microscopic mechanism and spectral structure under the specified geometry and coupling, in six steps: KMS locking of the detailed-balance ratio, branching geometry, the weak-coupling limit, optimality of the spectral gap, the three-timescale relaxation hierarchy, and the window of validity of the nonlinear layer; the full derivations are placed in Appendix B. Throughout, one odd prime q is fixed, serving at once as the local prime, the residue-field size, and the $(q + 1)$ -regular-tree parameter; numerical verification takes $q = 7$.

We first draw the logical boundary. The standard GL_1 BC system provides the reversible modular flow and its KMS equilibrium states; it does not by itself specify a GKSL dissipator. The provenance of the generators below is conditional: the device must simultaneously realize the Hecke adjacency of GL_2/PGL_2 , the half-line boundary given by Borel/valuation, and an

environment exchanging energy at that gap; under these conditions the Hecke Markov semigroup on the classical pointer algebra is canonical, and the quantum GKSL lift is realized by the chosen physical coarse-graining or bath dilation. The formulas of this section are therefore both design constraints and a falsifiable acceptance sheet for hardware; they do not constitute a dissipation theorem holding automatically for arbitrary BC representations. This is also the device-level form of the compilation boundary of §3.3.

The conclusions of the six steps are as follows.

- (1) The detailed-balance ratio is KMS-locked to $\Gamma_{\uparrow}/\Gamma_{\downarrow} = q^{-\beta}$ ((61)); the total rate κ is not constrained by the equilibrium structure and serves as the clock; κ is taken as the rate unit.
- (2) The candidate geometry is the Hecke adjacency of the Bruhat–Tits $(q+1)$ -regular tree ((62)); a tree step is the composite of additive translation and scaling, and the branching ratio automatically realizes the Boltzmann weight at $\beta = 1$.
- (3) Under finite truncation the hypotheses of Davies’ theorem hold item by item, the weak-coupling limit exists, and the secular generator is the Hecke Markov semigroup with jump ratio $q^{-\beta}$, together with its GKSL lift.
- (4) The radial relaxation gap is $\text{gap} = 1 - 2\sqrt{q}/(q+1)$ ((64)), attaining the Kesten–Alon–Boppana asymptotic bound within the comparison class of fixed-degree, uniform-edge-weight, reversible regular-graph walks.
- (5) The relaxation spectrum splits into three timescales ($q \geq 5$): the radial band at $O(1)\kappa$, primitive-character relaxation $\nu = \kappa/(q+1)$ ((66)), and the superselection gap $\Theta(q^{-K})$ ((67)).
- (6) The three-mode test window of the nonlinear mean-field layer is $gt \lesssim 0.5$, with the a priori truncation bound (69).

The cooling schedules and stopping thresholds of the solving and search instances, and the acceptance numerical twin, all rest on these three timescales (§5.1.3, §5.1.4, §8.4).

4.1.4 Control and Readout

The final link of a run is stopping and delivery. The arithmetic module’s control surface takes, per the control $u(t)$ of §3.2, cooling schedules, switching sequences, and conditional branches; the scheduling of stopping criteria follows §3.5. For low-temperature solving in finite systems, the stopping event is the target-sector occupation exceeding a pre-calibrated threshold. Glassy relaxation stagnation shows as insufficient target occupation, accompanied by growing relaxation times and aging scaling of response functions; unsatisfiable constraints keep the penalty energy at a positive floor. A finite observation window can only accumulate these two statistical signatures; it cannot elevate the distinction between “stalled” and “needs more time” into a general decision.

For a specified ergodic generator, when the constraints are satisfiable and the gap is positive, relative-entropy monotonicity makes fixed-temperature evolution converge to the corresponding Gibbs state; the low-temperature limit then concentrates the weight in the lowest-penalty sector. The readout on which feedback relies is QND measurement (§4.1.1); in the loop the register charges are constants of motion and control does not disturb storage (§3.8); QND-read conserved charges can directly select the next segment’s pumps and pinnings. A first machine

may implement the same switching law on an FPGA or a PC for calibration, logging, and compatibility; in such an implementation the FPGA or PC is only a control/measurement boundary and belongs neither to the machine's data format nor to its computational semantics.

On the state selected by τ_b , the arithmetic module outputs along the three readout forms of §3.4: evaluation through response functions, solving through target-sector occupation, statistical estimation through Born readout. What can be read out is limited by symmetry. The Galois group $\text{Gal}(\mathbb{Q}(\zeta_{|G|})/\mathbb{Q})$ acts on the character group by $\chi \mapsto \chi^t$; the complex-conjugation element $t = -1$ induces at the operator level the linear $*$ -automorphism $\sigma : a_\chi \mapsto a_{\bar{\chi}}$ (realized by conjugation with the group automorphism $g \mapsto g^{-1}$; from $a_\chi^* = a_{\bar{\chi}}$ one gets $\sigma(a^*) = \sigma(a)^*$, so σ is compatible with $*$). In the symmetric measurement model of this paper, in which readout does not distinguish Galois conjugates, reportable real-valued criteria must be simultaneously self-adjoint and σ -invariant; the latter expresses the indistinguishability of Galois conjugates. Among the linear combinations $c_1 a_\chi + c_2 a_{\bar{\chi}}$ spanned by a conjugate orbit $\{\chi, \bar{\chi}\}$, σ -invariance requires $c_1 = c_2$ and self-adjointness requires $c_2 = \bar{c}_1$; their intersection is $a_\chi + a_{\bar{\chi}}$ with real coefficients. In particular $i(a_\chi - a_{\bar{\chi}})$ is self-adjoint but flips sign under σ , hence unmeasurable. Higher combinations fall in the self-adjoint part of the σ -fixed commutative algebra, generated by real-coefficient combinations of $a_\chi + a_{\bar{\chi}}$, $a_\chi a_{\bar{\chi}}$, and occupation-type operators; the list is closed. Quantities breaking conjugation symmetry fail one of the two criteria and do not exist at the measurement level. This condition restricts the set of readable operators, but it does not exclude noise changing the expectations of these invariant operators; the corresponding protection scope is distinguished by noise type in §6.9. The reportable quantities of each instance are bounded by this list.

Output transducers turn the single record r into an action, a mode excitation of another finite physical system, or compatible display and file formats; the $\mathcal{R}[b; \rho]$ formed by repeated runs serves statistical estimation and audit, and thereby forms the external condition s_{k+1} . The choice of the next boundary belongs to control and scheduling; it is not prescribed by the output port alone.

4.2 The Interaction Module

The four links of the arithmetic module are now complete; the interaction module takes over the machine's boundary with the world. Its defining feature is that the environment is a participant: the parts' inputs and outputs are physical conditions themselves, and every run starts in the world and ends in the world; both ends of the loop (7) therefore land on s_{phys} , and what is realized here is the perception-action loop listed in §2.7. The interaction module has no state space of its own: the occupations and phases written by transduction are the arithmetic module's data forms, constraint increments enter the boundary, and actuation relies on the storage-compatible QND readout; the two modules share one carrier and differ in the orientation of the boundary.

4.2.1 Transduction

Transduction maps locally distinguishable features of external fields into the machine's boundary and state increments. Optical, acoustic, mechanical, thermal, or chemical fields enter the run through (8), along four channel types: boundary reconfiguration, occupation increments ΔN , phase increments $\Delta\varphi$, and constraint energy; the i -th feature enters with likelihood factor L_i , its constraint energy being

$$E_i(\mathbf{v}) = -\log L_i(\mathbf{v}). \quad (42)$$

Visual tasks may take likelihood factors of phase delay, frequency band, motion, occlusion, or object hypotheses; acoustic tasks may take those of harmonics, phase, direction, and sources. The correspondence between features and modes is fixed jointly by sensor physics, task model, and calibration; the correspondence table belongs to the declared part of the boundary program π , and its calibration cost is charged to the resource vector (§3.9). The transduction bandwidth B_{tr} is given by coupling rates and mode linewidths and is one of the device-side sources of the tracking condition (53); in the weak-signal window transduction is count accumulation, whose shot-noise floor is precisely the Poisson statistics of the detection instance (§5.2.1).

4.2.2 Evidence

When features are statistically conditionally independent and written into disjoint mode sets, evidence composes by energy:

$$L(\mathbf{v}) = \prod_i L_i(\mathbf{v}) \iff E(\mathbf{v}) = \sum_i E_i(\mathbf{v}). \quad (43)$$

That independent composition multiplies and energy adds takes this form on the interaction side; the Gibbs posterior of the inference instance follows from it (§5.3.1). Evidence energy entering the boundary is precisely a pinning share, the same mechanism as compilation’s penalty energy but of different origin (§3.3): penalty energy is written in by the program’s declaration, evidence energy by the world at run time, both as soft constraints. Multiplicativity presupposes conditional independence: correlated observations of overlapping views must be modeled explicitly, and correlated likelihoods may not be multiplied directly.

4.2.3 Actuation

The actuation port turns the action-relevant readouts of a single record r , through admissible feedback relations, into new boundary conditions; actuators change external fields, and the new physical conditions return the corresponding $\Delta N, \Delta \varphi, \Delta Q[b]$. The loop is written

$$s_{\text{phys}} \longrightarrow (b(t), Q[b]) \longrightarrow \rho_* \longrightarrow (r, \rho^+) \longrightarrow \text{actuator} \longrightarrow s'_{\text{phys}}. \quad (44)$$

The admissible feedback relation is precisely the storage-compatible readout: feedback relies on QND records, and the loop does not disturb retained states (§4.1.4). Actuation resolution and measurement shot noise jointly constitute the loop’s noise floor, the device-side origin of ξ_k in the servo instance (§5.2.2); the actuation timescale T_{act} enters the loop period ((54)).

4.2.4 Inter-Machine Coupling and Legacy Interfaces

Two machines may couple mode degrees of freedom of the same type, transferring occupations, phase relations, and constraint factors; conservation matching on shared modes (compatibility of conserved-charge flows at the junction) and timing hold by the interface conditions of the composition semantics (§3.8). Stable conserved charges, phase relations, and constraint events provide the retainable states within the system, with the corresponding variables readable by probe response. Facing legacy networks, displays, or audit archives, the boundary layer further converts these variables into numerals, images, audio, or files.

This interaction front end creates no new dynamical relations; it prescribes how physical conditions enter and leave the machine.

5 Running Instances

In this section the parts of the two modules combine into instances, unfolded in three groups: the arithmetic module running alone, the interaction module running alone, and the two modules running in composition. Within each group the instances progress by complexity, and each walks the full chain of task, program, completion and correctness, and resources in the semantics of §3; the three dilemmas of §1.1 close at the corresponding instances.

5.1 Instances of the Arithmetic Module

When the arithmetic module runs alone, the environmental share is controlled to laboratory conditions. Four instances progress by complexity: the elementary operations give the minimal runs; evaluation, solving, and search give the three loads.

5.1.1 Elementary-Operation Instances

The two elementary operations, multiplication and addition, are directly the minimal runs. The multiplication task gives n, m and requires the output nm : the program π is the joint declaration of occupation writing, multiplicative isometry, and occupation readout, (n, m) is the input x , and the run takes the coherent mode.

Step 1, preparation. Occupation writing gives $|n\rangle$.

Step 2, evolution. Apply μ_m ; in the subsequent evolution the occupations are constants of motion ((31)), multiplication requires no stepwise execution, and adding occupations is the result: $\mathbf{v}_\Lambda(nm) = \mathbf{v}_\Lambda(n) + \mathbf{v}_\Lambda(m)$.

Step 3, readout. QND readout returns $\mathbf{v}_\Lambda(nm)$; completion is event completion, and correctness is decided on the single record by unique factorization (§3.5, §3.4).

Addition runs in the phase sector mod q . The task gives residues $a, c \in \mathbb{Z}/q\mathbb{Z}$ and requires the output $a + c$; addition is assembled from scalings and translations (§4.1.2, Step 4), and the ideal operator relations are exact.

Step 1, preparation. The initial state is $F_q|a\rangle$.

Step 2, evolution. Apply the phase class V_c ; by (6) this yields $F_q|a + c\rangle$.

Step 3, readout. Interfere with a reference phase for the q -point decision; completion is event completion, correctness is decided on the single record (as for multiplication), and device error is charged to the phase-precision entry of the resource vector (§3.9).

The two micro-runs give the native pattern of the elementary operations: multiplication goes by occupation composition, addition by phase assembly, with resources charged to the resource vector (§3.9).

5.1.2 The Evaluation Instance

The evaluation task gives parameters (σ, t) or a character χ and requires as output the values of analytic functions on vertical lines; the program π is the joint declaration of probe preparation, evolution window, and response measurement, the parameters are the input x , and the run takes the coherent mode; this instance invokes coherent superposition, one evolution of the probe state carrying the joint phase of all integers.

Step 1, preparation. Take the Gibbs coherent state $|\psi_\sigma\rangle = \zeta(2\sigma)^{-1/2} \sum_n n^{-\sigma} |n\rangle$ ($\sigma > 1/2$); the probe is a product state (§4.1.1), with linear preparation cost.

Step 2, evolution. The probe evolves freely in the evolution window, $e^{iHt}|n\rangle = n^{it}|n\rangle$.

Step 3, measurement and readout. The response amplitude is

$$\langle \psi_\sigma | e^{iHt} | \psi_\sigma \rangle = \frac{1}{\zeta(2\sigma)} \sum_n n^{-2\sigma+it} = \frac{\zeta(2\sigma - it)}{\zeta(2\sigma)}, \quad (45)$$

the machine's autocorrelation function being verbatim the value of ζ on a vertical line; inserting U_χ one-sidedly at readout, the cross-correlation gives, under the same evolution, $\langle \psi_\sigma | U_\chi e^{iHt} | \psi_\sigma \rangle = L(2\sigma - it, \chi) / \zeta(2\sigma)$. Output precision is controlled by the shot noise $M^{-1/2}$ of the sample count; completion is statistical completion (§3.5), correctness is judged as statistical correctness (§3.4), and preparation, evolution window, and sample count are charged to the resource vector (§3.9).

The readout port also yields verifiable instances of algebraic, spectral, and positivity relations; their proof semantics appears in §7.

5.1.3 The Solving Instance

The solving task gives a constraint family and requires as output a configuration satisfying it; compilation maps the constraints to penalty energies per §3.3, the run takes the relaxation mode and raises β along a cooling law [31], the target-sector probability is controlled by the penalty gap and the degeneracy, and readout carries algebraic verification.

Constraint tasks are stated by the specification of §3.1, the statement layer consisting of Diophantine conditions and conservation requirements. The DPRM theorem shows that every recursively enumerable set is representable as the existential projection of some Diophantine equation [6]; this establishes the expressive power of the constraint language, not the decision time of a finite device nor a few-body physical compilation. For a fixed witness bound, equations can be turned into finite penalty energies; general polynomials contain cross-variable products, and their realization as few-body coupling networks, with the gap cost, must be supplied separately. Unbounded recursively enumerable semantics belongs to machine families with stepwise-growing truncations; a single finite machine does not carry that semantics.

We now run a binary-occupation subset-product task in full: given the prime set $S = \{2, 3, 5, 7, 11\}$ and target $N = 330$, find $v_p \in \{0, 1\}$ with $\prod_{p \in S} p^{v_p} = N$. The compilation stage uses only S and $\log N$, without precomputing $v_p(N)$. The penalty energy is

$$H_{\text{pin}}(\mathbf{v}) = \left(\sum_{p \in S} v_p \log p - \log N \right)^2, \quad v_p \in \{0, 1\}. \quad (46)$$

Within the restricted mode set S and the binary constraint $v_p \in \{0, 1\}$, unique factorization guarantees that $H_{\text{pin}} = 0$ if and only if the occupation vector gives the target subset; the identity of the zero space is thus decided by the input N , yet no answer was written in at compile time. In the fourfold split of §3.2, S , the proposal kernel, the cooling law, and the stopping threshold constitute the reusable program π , and $\log N$ is the present input x . This penalty couples channels, so the generator is in general no longer a single-channel tensor sum, and its gap must be computed on the joint network.

Step 1, preparation. The binary modes are initialized at finite high temperature, and the boundary opens a multi-step proposal kernel covering the required sector. The proposal set must generate the whole finite configuration graph; (47) (see the search instance) by itself provides no ergodicity proof.

Step 2, running. The temperature schedule is set from the measured or computed gap of the joint generator. At fixed temperature, if the generator is ergodic and satisfies detailed balance,

the state converges to $e^{-\beta H_{\text{pin}}}/Z$; as β grows, the weight concentrates on the lowest-penalty sector at scale $e^{-\beta\Delta}$, with Δ the minimal positive penalty gap. In a finite system this process is low-temperature concentration, not a non-analytic phase transition.

Step 3, stopping and readout. When the target-sector occupation exceeds a preset threshold, sampling is triggered; QND readout returns a candidate $\mathbf{v}$, and the multiplicative side verifies $\prod_p p^{v_p} = N$ directly. Verification failure means continuing the run or restarting, forming a Las Vegas protocol: completion is event completion (§3.5), and correctness is decided on the single record by the multiplicative check (§3.4). For the present input the zero-penalty candidate is $(1, 1, 1, 0, 1)$, read out as $\{2, 3, 5, 11\}$; this vector is a run result rather than a compile input.

If N is not in the binary multiplicative span of S , the finite configuration set gives a strictly positive floor $\Delta_0 = \min_{\mathbf{v}} H_{\text{pin}}(\mathbf{v}) > 0$. Finite observation can only report that no zero-penalty candidate has appeared, together with the current minimal penalty; without an exhaustive certificate, a low-temperature miss does not by itself decide unsatisfiability.

This instance also marks the decomposability boundary. If a polynomial-time compiler turned arbitrary instances into $\sum_p h_p(v_p)$, channelwise minimization would already deliver the solution at the compile layer; nontrivial constraints that do not pre-write answers typically appear as cross-channel penalties, possibly with small gaps and glassiness. The numerical $N = 392$ instance of §8.4 tests exactly the quench and restart cost of this honest encoding; its time, energy, and restart overheads are charged to the resource vector (§3.9).

In both the evaluation and solving instances the statistics and thermalization are native resources: pseudorandom numbers and sampling reconstruction do not appear, and Dilemma two of §1.1 closes here.

5.1.4 The Search Instance

The search task requires locating an element satisfying the constraints in configuration space; the program π is an alternating schedule of additive mixing and multiplicative verification, and the run is organized by the monitored and feedback modes; additive mixing carries the search, multiplicative structure carries the evaluation.

Step 1, assembling the mixing kernel. The mixing kernel is translations $S^{\pm q}$ sampled randomly by a spectral shape, assembled from V and F .

Step 2, running. Mixing and verification alternate; the ergodicity of the search is given by coprimality. Let $r \nmid q$ be prime; if $r \mid n$ and $r \mid (n \pm q)$ then r divides their difference q , a contradiction; hence

$$v_r(n) v_r(n \pm q) = 0 \quad (r \nmid q) \quad (47)$$

holds pointwise. Under natural density $\mathbb{P}(v_r = a) = (1 - r^{-1})r^{-a}$. Given $v_r(n) \geq 1$, the event $v_r(n \pm q) = 0$ is certain; given $v_r(n) = 0$, for $b \geq 1$,

$$\mathbb{P}(v_r(n \pm q) = b \mid v_r(n) = 0) = r^{-b}, \quad \mathbb{P}(v_r(n \pm q) = 0 \mid v_r(n) = 0) = \frac{r-2}{r-1}. \quad (48)$$

From $\mathbb{E}v_r = 1/(r-1)$, $\text{Var}(v_r) = r/(r-1)^2$, and $v_r(n)v_r(n \pm q) = 0$, the one-step valuation autocorrelation coefficient is $-1/r$.

Step 3, stopping and readout. A verification hit is event completion, and correctness is decided on the single record (§3.5, §3.4); mixing steps and verification counts are charged to the resource vector (§3.9).

The ergodicity paragraph describes only the marginal correlation of a single valuation under counting measure; it does not yield the gap or the mixing time of the whole additive Markov

chain. A fixed single step q moreover preserves $n \bmod q$, making the chain reducible. The mixing rate of a multi-step Metropolis kernel [17] must be analyzed separately for the target distribution, the step distribution, and the boundary conditions; the results of §8.6 are finite numerical benchmarks.

5.2 Instances of the Interaction Module

When the interaction module runs alone, content processing takes its simplest form. Detection gives the minimal run with the world; servo and stabilization give closed-loop control.

5.2.1 The Detection Instance

Detection is the interaction module’s minimal run. The task gives one distinguishable feature and a threshold Θ .

Step 1, transduction. $\mathcal{T}_\Lambda$ maps the feature strength to a pump increment on the target mode.

Step 2, accumulation and monitoring. Evolution is count accumulation, and the monitoring criterion is occupation crossing the threshold, $N \geq \Theta$. In the weak-signal window the counts are approximately Poisson: with background rate λ_0 and signal rate $\lambda_1 > \lambda_0$, the discrimination error between the two hypotheses over an observation window T decays exponentially in T , with exponent given by the Poisson large-deviation rate; the information rate is

$$D(\lambda_1 \parallel \lambda_0) = \lambda_1 \log \frac{\lambda_1}{\lambda_0} - \lambda_1 + \lambda_0. \quad (49)$$

Step 3, stopping and readout. Threshold crossing triggers stopping, and QND readout reports presence or absence; completion is event completion, and correctness is decided on the single record (§3.5, §3.4). The threshold Θ apportions the two exponents between false alarm and miss; observation window and pump power are charged to the resource vector (§3.9).

5.2.2 The Servo and Stabilization Instances

Servo closes a single loop. The task gives an environmental variable y in a normed space (d the induced metric) and a target set Y_{acc} .

Step 1, transduced writing. Transduction maps y_k into boundary and occupation increments.

Step 2, decision and actuation. The action-relevant part of the readout is turned by the actuation port into a new boundary condition, the actuator changes the external field, and the new physical condition returns through transduction ((44)); the loop relies on storage-compatible readout (§3.8). One circuit of the loop induces from the run law (18) a map on the environmental variable, $y_{k+1} = F(y_k) + \xi_k$, where F is the composite of transduction, decision, and actuation, and ξ_k is given by measurement shot noise and actuation resolution.

Step 3, convergence. If F contracts with gain $\theta < 1$ to a fixed point $y_* \in Y_{\text{acc}}$, then

$$d(y_k, y_*) \leq \theta^k d(y_0, y_*) + \frac{\sup_k \|\xi_k\|}{1 - \theta}, \quad (50)$$

convergence is geometric in loop count with the steady-state error given by the noise floor; completion is closed-loop completion, correctness is $d(y_k, Y_{\text{acc}})$ entering the prescribed radius (§3.4), and transduction, actuation, and loop period are charged to the resource vector (§3.9).

Stabilization hands control to structure.

Step 1, compilation. The program compiles the target behavior into a stable state or limit cycle of the constrained dynamics; control only imposes the boundary.

Step 2, dwell and recovery. Recovery from perturbations is carried by relaxation, the recovery timescale being the inverse of the corresponding gap; dwell within the attraction basin is given by the barrier through the Arrhenius escape rate $r_{\text{esc}} = \kappa_0 e^{-\beta \Delta_B}$ (Δ_B the minimal escape barrier).

Step 3, stabilization-completion. The dwell probability over a window $[0, T]$ satisfies $P_{\text{stay}} \geq 1 - T r_{\text{esc}}$, and the stabilization-completion criterion is $T r_{\text{esc}} \ll 1$ (§3.5); correctness is the environmental variable dwelling in the target set (§3.4), and boundary maintenance and recovery times are charged to the resource vector (§3.9).

The passive walker of §1.1 here takes its general form: gait is an eigenmode of structure, balance is carried by the stable state, control pays only the boundary-maintenance entry, and millisecond-by-millisecond re-solved optimization does not appear.

5.3 Composite Instances

Composition of the two modules takes three forms: same-domain composition superposes transduction shares and pinning shares into one boundary, time-division composition uses registers to carry inter-segment transfer, and spatial composition couples two machines by shared modes. Time division and space implement the serial and parallel composition of §3.8; same-domain composition is the superposition of boundary families within a single run, involving no multi-computation composition, and its legality is precisely that the union of the joint constraints is realizable and conservation matching holds, the single-run form of the interface conditions of §3.8. Below we give, in order, the same-domain form, the engineering forms of time division and space, the complete time-division loop, and close with physical simulation and the full closed loop.

5.3.1 The Inference Instance

Inference is the first composite instance, of the same-domain kind: evidence enters the boundary by transduction (interaction module), the posterior is produced by thermalization (arithmetic module), and a single run belongs to both boundary families. The task gives a prior and multi-channel sensor evidence and requires reporting the posterior over configurations or its maximum a posteriori estimate; the program π is the declaration of prior pinning, the evidence correspondence table, and the sampling schedule. Hypotheses are represented by configurations $\mathbf{v}$.

Step 1, writing. The prior enters the boundary as pinning energy $E_0(\mathbf{v}) = -\log P_0(\mathbf{v})$; the i -th independent sensor evidence enters the same configuration through transduction as constraint energy E_i ((42)).

Step 2, thermalization. By the additivity of evidence (43), the Gibbs weight at $\beta = 1$ gives

$$P(\mathbf{v}) = \frac{e^{-E_0(\mathbf{v}) - \sum_i E_i(\mathbf{v})}}{Z} = \frac{P_0(\mathbf{v}) \prod_i L_i(\mathbf{v})}{Z}, \quad (51)$$

which is the Bayes posterior.

Step 3, sampling and readout. Posterior sampling is the solving instance run at $\beta = 1$, and maximum a posteriori estimation is the solving run at increasing β ; estimation precision is controlled by the shot noise $M^{-1/2}$ of the sample count, completion is statistical completion (§3.5), correctness is statistical correctness, and sample count and transduction bandwidth are

charged to the resource vector (§3.9). Inference thus sits in the intersection of the two modules' boundary families.

5.3.2 Sequential Programs and Composition

The two engineering forms of composition appear at this level: time-division composition bridges segments by boundary switching and registers, and spatial composition couples machines by shared channels; same-domain composition has already appeared in the inference instance (§5.3.1). A single anneal gives one act of solving; multi-step computation, by the serial mechanism of §3.8, feeds the actual record r_k and the post-measurement state $\rho_{k,+}$ of (7) into the next segment's boundary b_{k+1} : QND readout takes the N_p charges (§4.1.1), registers take decoupled character sectors, and conditional branching takes pump reconfiguration (§4.1.4).

Step 1, inter-segment transfer. The low-temperature candidate of stage k is QND-read and written into the register sector R_k ; the pinning target v^* of stage $k+1$ is set by the readout value of R_k . A first machine may let a classical controller temporarily stand in for this feedback law, but that controller neither stores nor interprets the machine's arithmetic state; the feedback basis remains QND measurement, and the loop does not disturb storage (§4.1.4).

Step 2, feedback programs and the instruction table. The native declarative mode gives only constraints, conserved charges, and pump schedules; its sequentiality is organized by post-readout pump switching. For interoperation with existing algorithms, the imperative mode's instruction table is also defined,

$$\{\mu_q^\pm, U_\chi/V_c, F, \text{QND}(N_p) \text{ branch, anneal segment, register read/write}\},$$

whose finite sequences compile to the same physical feedback law. By Step 1, the two modes compose closely under register semantics; the program-level error budget is the union of the per-segment errors, valid under register semantics: QND readout and the loop's non-disturbance of storage exclude inter-segment amplification (§3.10).

Step 3, finite-tape simulation. Take L addressable prime modes to represent a tape of length at most L , with an additional head position and a finite control state. One Turing step consists of an occupation read/write on the mode under the head, one QND conditional branch, a head update, and a finite-state update. Local read/writes are done by finitely many $\mu_{p_i}^\pm$ operations; the total cost also includes the addressing time of selecting the i -th mode. Independent spatial addressing, frequency multiplexing, and compressed encoding carry respectively the space, resolution, and controller costs given in §6.6, so arbitrary mode access cannot be booked uniformly at unit cost.

Step 4, two compilation modes. The imperative mode organizes QND readout, mode addressing, and pump switching by an explicit finite controller; for fixed L it simulates Turing machines of tape length at most L with finite resources. Unbounded universality belongs to the uniform machine family of growing L , the controller being part of the simulation. The declarative mode compiles constraints within a fixed witness bound into penalty energies and anneal segments; DPRM guarantees only language-level expressive power. Both modes must charge addressing, control, and gap costs to the resource account; prime-mode encoding by itself does not confer constant-cost universal computation.

The arithmetic semantics of parallel and coupled composition has two levels, landing by the conditions of §3.8. Independent composition: the joint state space is the tensor product, the joint partition function is the product, and $\log Z$ and energy are additive. Coupled composition: when the two modules share a channel set C , the joint constraint is the union of the two

sides' constraints plus conservation matching on C (the upstream module's output charge is the downstream module's input charge), realized as joint selection rules on the shared channels, with unshared channels keeping their own pumps. The admissible-state set after composition is still convex; when energy and constraints are feasible, the maximum-entropy variation still gives a unique equilibrium state. The joint dynamics depends further on the geometry, edge weights, and bath couplings of the shared channels and cannot be determined by the union of the constraint sets alone; tensor-product semantics is the special case $C = \emptyset$. The price of composition is computable: a shared channel's charge is no longer a local conserved quantity of a single module, and its sector retention time is recomputed from the joint network's relaxation spectrum (32). Composability is precisely the machine form of the third basic fact (independent composition multiplies).

The working form of spatial composition is the coupling of a front end to a computing machine. An acoustic front end writes likelihood factors of harmonics, direction, and sources as constraint energies through shared modes ((42)); the computing machine runs solving on them for source separation, and the readout is delivered through actuation ports or legacy interfaces; conservation matching and timing synchronization on the shared channels hold by the interface conditions of §3.8.

5.3.3 The Identify–Decompose–Act Instance

The complete form of time-division composition consists of three boundary segments. The task gives a superposed physical quantity in the environment and corresponding action rules; the three segments share one machine, the boundary switches by segment, and registers carry the inter-segment transfer (§3.8).

Step 1, identify. The interaction module transduces the target quantity, obtaining an estimate of $\log N$ with its evidence weight, written into the register.

Step 2, decompose. The arithmetic module runs subset-product solving ((46)) on the register data, obtaining the composition $\mathbf{v}$, verified multiplicatively.

Step 3, act. By $\mathbf{v}$ an actuation channel is selected, and the action changes the external conditions.

End-to-end correctness is given by the inter-segment interfaces. Let the operational distances of the identify, decompose, and act segments be $\varepsilon_1, \varepsilon_2, \varepsilon_3$, with each segment's output falling in the next segment's reachable input class (the interface conditions of §3.8); then the distance of the composite run law from the task requirement satisfies

$$\varepsilon_{\text{tot}} \leq \varepsilon_1 + \varepsilon_2 + \varepsilon_3, \quad (52)$$

by the triangle inequality of the operational distance and the monotonicity of distances under channel action, recursively segment by segment. Completion takes the last segment's criterion, event or closed-loop (§3.5); the error budget is allocated by segment, and each segment's resources are merged and charged to the resource vector (§3.9).

5.3.4 The Physical-Simulation Instance

The simulation task gives a target system in the thesis class (excitations countable, energy additive, independent composition multiplicative; §3.12) and requires estimating its equilibrium observables, spectral functions, or evolution statistics.

Step 1, boundary matching. The program π is a declaration of boundary matching: the mode set takes the target’s excitation spectrum, the admissible couplings assemble the target’s interactions, and thermal contact takes the target’s temperature; compilation generates conditions, not trajectories (§3.3).

Step 2, running. The preceding instances hereby become the components of simulation: equilibrium estimation is the solving instance, spectral functions the evaluation instance, ground-state search the search instance, and multi-sector targets are organized by sequential programs and composition. The evolution itself needs no replay: the target’s state space is carried by the machine’s state space, the dynamics is selected by the boundary, and the storage and stepwise update of exponentially many amplitudes do not appear; Dilemma one of §1.1 closes here.

Step 3, scope and accounting. The scope of simulation is drawn by resource accounting: when the target generator falls in the boundary family, simulation is native; the excess is compiled into penalty energies and assemblies, with the cost charged to the resource vector (§3.9); writing in target transitions rate by rate is protocol emulation, outside native simulation (§3.3). Completion and correctness take the criteria inherited from the component instances invoked (§3.5, §3.4).

5.3.5 The Sustained-Tracking Instance

The tracking task gives a time-varying quantity and a target set in an open environment: the environmental state varies slowly, the machine reports persistently over rolling windows, and no terminal state is required (§3.5). The program π is the declaration of a four-step schedule (the transduction declaration, cross-loop reuse of the inference and solving programs, and the actuation rules). Take the object of the identify–decompose–act instance (§5.3.3) as the running example: the superposed physical quantity now varies slowly in time as $N(t)$ (e.g., the composition of a mixed signal switching), and one circuit of the loop passes through the four steps.

Step 1, transduction. The interaction module writes the environmental variables into boundary and occupation increments, taking time T_{tr} ; in the running example, the current harmonic and intensity features are written into evidence energy as likelihood factors ((42)).

Step 2, inference. Thermalization on the evidence energy gives the current posterior (§5.3.1), taking time T_{inf} ; in the running example, the current posterior of $\log N(t)$, with the previous circuit’s posterior pinned as this circuit’s prior.

Step 3, solving. Solving on the posterior or the constraints gives action candidates (§5.1.3), taking time T_{sol} ; in the running example, subset-product solving of the current composition $\mathbf{v}(t)$ with multiplicative verification.

Step 4, actuation. The actuation channel changes the external conditions and the new physical conditions re-enter transduction, taking time T_{act} ; in the running example, an actuation channel is selected by $\mathbf{v}(t)$, and by the next circuit the environment has changed.

Completion takes the rolling form of closed-loop completion: sustained tracking is the completed state; correctness is the environmental variable persistently falling within the prescribed radius in each rolling window (§3.4).

That tracking holds is a timescale condition: the environmental correlation bandwidth B_{env} must fall within the transduction bandwidth and the loop relaxation rate,

$$B_{\text{env}} \lesssim \min(B_{\text{tr}}, \kappa_{\text{loop}}), \quad (53)$$

where κ_{loop} is the loop relaxation rate; in the running example B_{env} is the switching bandwidth of the composition. Loads violating the condition belong to the high-speed servo limit of §8.7. Multi-sensor and multi-actuator loops are organized by the composition semantics: independent loops compose as tensor products, loops sharing environmental variables must satisfy the interface conditions of §3.8 and recompute the joint relaxation spectrum; inter-machine coupling transfers occupations, phase relations, and constraint factors. The limits of strongly coupled and glassy loads are handled by the gap and boundary criteria (§8.7).

When the three segments identify–decompose–act enter one loop, they take precisely the four-step form above (identify unfolds into the transduction and inference steps), and the time of content processing enters the loop period:

$$T_{\text{loop}} = T_{\text{tr}} + T_{\text{inf}} + T_{\text{sol}} + T_{\text{act}}, \quad B_{\text{env}} \lesssim \min(B_{\text{tr}}, \kappa_{\text{loop}}, T_{\text{loop}}^{-1}), \quad (54)$$

whereby the tracking condition (53) takes its composite form; a loop too slow at content processing loses tracking, a limit that appears only in composite running.

At this point Dilemma three closes. Perception and actuation are carried by the physical coupling of transducers and actuators, the object of decision is the physical state itself, and the world’s symbolic replica with its synchronization bandwidth no longer appears; what remains is the physical bandwidth of boundary transduction, drawn by the device’s transduction and relaxation timescales, whose device realization and acceptance appear in §8.

The three dilemmas close simultaneously within the same loop: content processing is carried directly by the state space, and step-by-step replay does not appear (§5.3.4); statistics and thermalization are native resources (§5.1.2–§5.1.3); the interface to the world is carried by physical coupling (this instance). Computation and perception–action are one within a single run, and the arithmetic workspace and the perception–action loop are given by the mix of calls on the same loop (§4).

6 Machine Properties

This section derives the machine’s derived properties from the computational semantics of §3 and the modules and instances of §4–§5; the general definitions are not repeated, and only their concrete manifestations and quantitative limits in the arithmetic workspace are given. The ten subsections state, in order: information metrics, addressing structure, and declarative programming characterize the structural differences from the place-value paradigm; storage, operations, input–output, and control give part specifications; native limits, the error model, and system-level limits draw the capability boundary; the section closes with a comparison against the prevailing paradigms.

6.1 Information Metrics

The machine’s information metric is carried by the energy spectrum: states are the basis vectors $|n\rangle$, the energy $\log n$ serves at once as address and as information content, and the metric and the dynamics share one reading. Three properties follow.

First, energy is capacity. The basis vectors with energy at most E are exactly $\{|n\rangle : \log n \leq E\}$, with count

$$\#\{n : \log n \leq E\} = \lfloor e^E \rfloor, \quad \log \lfloor e^E \rfloor = E + O(e^{-E}), \quad (55)$$

so the maximal encodable information in the energy window $[0, E]$ is E (natural units): capacity and energy are the same reading on the spectrum of $H = \log n$. Unit conversion is given by the lightest mode: a single occupation quantum of mode 2 carries energy $\log 2$, and its energy window distinguishes exactly the two states $\{|1\rangle, |2\rangle\}$: one bit is one quantum of the lightest prime mode, and a quantum of mode p carries $\log_2 p$ bits. A Gibbs state's entropy at given mean energy can exceed that energy, reflecting the distribution's energy tail; (55) is a capacity statement under an energy-cap constraint.

Second, source coding. Coding takes the “minimal address energy” form: let the symbols X to be stored occur with distribution P , a code be an injection c into basis labels, and the address energy paid at readout be $\log c(X)$. Coding the i -th symbol in probability-descending order as $|i\rangle$ gives $P_i \leq 1/i$, so

$$\mathbb{E}[\log c(X)] = \sum_i P_i \log i \leq \sum_i P_i \log \frac{1}{P_i} = H(P); \quad (56)$$

conversely, every injective code satisfies $\mathbb{E}[\log c(X)] \geq H(P) - \log(H(P) + 1) - \log e$ [5]. The optimal address energy therefore lies within an $O(\log H)$ neighborhood of the Shannon entropy [20]: compression is placing high-probability symbols at low-energy addresses. The maxim of §6.2, that the number is the address and the factor structure is the content, hereby acquires metric form: the energy of an address is its information content.

Third, two ledgers kept apart. The address energy $\log n$ is the configurational energy of the encoding state itself, carried back and forth by writing and readout; the erasure heat (34) is the entropy cost of resetting an unknown distribution. The former is carried reversibly, the latter paid irreversibly; merging the two double-counts the storage cost. Noiseless channels transferring occupations across the boundary are bounded by the same capacity identity; channel capacity under thermal noise belongs to future work.

6.2 Addressing Structure

The label of a state-space basis vector is the number itself: the number is the address, the factor structure is the content, and addressing is carried directly by the state-space structure. In place-value machines a large share of adder throughput is spent on addressing: program-counter increments, base-plus-offset, array indexing. Set against the four duties of addition, this machine's carriage and diversion of those duties constitute the first property; comparison and content addressing follow.

First, the diversion of addition's duties.

(1) Addressing and traversal: the coprimality identity (47) gives the one-step anticorrelation of a single valuation; it neither implies independence of the full address vector nor guarantees ergodicity of the proposal chain; addressing and traversal are carried by concrete multi-step kernels and their boundary rules (§5.1.4).

(2) Control flow: loop counting and jumps consume a significant share of place-value adder throughput; this machine's control is carried by temperature and pump sequences (§4.1.4), and the program counter has no counterpart.

(3) Accumulation: the macroscopic occupation of sample statistics and order parameters is realized at the measurement end by physical ensemble averaging; the detector count is the integrator, and summation completes at the readout layer.

(4) Arithmetic addition: only this one passes through the F assembly (§4.1.2, Step 4), and the first three have already absorbed a large share of the addition load of place-value machines.

Second, magnitude comparison. Comparison splits into two resolution regimes. Coarse comparison ($|\log m - \log n| = \Omega(1)$): $H = \log n$ is a global observable on the additive side, order is spectral order, a single QND energy readout completes it, and overflow is spectral out-of-range monitoring. Exact comparison: the energy difference of adjacent numbers, $\log(1 + 1/n) \sim 1/n$, shrinks exponentially with scale, so analog readout is not viable; instead the feedback program layer (§5.3.2) performs mixed-radix conversion: modulus-by-modulus elimination, each step costing $O(1)$ modular additions and one QND branch, the number of steps set by the chosen modulus cover and the reconstruction algorithm; if the modulus product exceeds the dynamic range, standard CRT reconstruction has polynomial cost, and no exponentially precise analog energy readout is needed. The classical pain point of residue number systems [35] (comparison and overflow costing nearly a full reconstruction) is thereby taken over by this two-tier division of labor: the coarse tier is a global address metric outside the residue representation (with no place-value counterpart), and the exact tier is of the same order as place value.

Third, content addressing. Constraints pin the input content, low-temperature readout returns candidate addresses (§5.1.3), the machine addresses by content, and the address/content dichotomy of the von Neumann architecture does not appear in the state space. The machine is hence self-sufficient in *computational semantics*: declarative low-temperature loads use no program counter; the imperative mode does require explicit finite-state control, heads, and mode addressing. If an external controller takes on those duties, it must be part of the resource model and cannot be omitted from the running account.

6.3 Declarative Programming and Native Logic

This machine’s programs live on the boundary: declaring constraints and conserved charges completes programming (§3.2); logic is natively carried by the divisibility lattice. Two properties unfold.

First, declarative programming. The maximum-entropy variation (41) makes the equilibrium state under feasible constraints unique; the run path still requires the generator to be specified separately. Two machines have the same evolution operator only when constraints, jump geometry, edge weights, bath couplings, and timescales all coincide. Multi-segment computation is organized by QND–pump feedback loops; the controller and its addressing cost are part of the imperative mode’s resource account.

Second, native logic. The propositional lattice is the divisibility lattice $(\mathbb{N}^*, |)$, with gcd as meet and lcm as join; the distributive law

$$\gcd(a, \text{lcm}(b, c)) = \text{lcm}(\gcd(a, b), \gcd(a, c))$$

follows prime by prime from

$$\min(x, \max(y, z)) = \max(\min(x, y), \min(x, z)).$$

Complementation would require some a' with $\gcd(a, a') = 1$ and $\text{lcm}(a, a')$ the top of the lattice, but $(\mathbb{N}^*, |)$ has no top, so the lattice is uncomplemented. Existential quantifiers over Diophantine constraints are realized by low-temperature candidates within finite bounds plus algebraic verification. DPRM [6] shows that the unbounded constraint language expresses all recursively enumerable sets; a single finite device realizes only fixed modes and witness truncations, negative conclusions further require exhaustion or independent certificates, and the parity barrier in the sieve interface is one concrete limitation of this certificate semantics (Appendix C.1).

6.4 Storage Properties

Storage is carried by conserved charges: data are written into the occupation numbers N_p of decoupled sectors, and the register is the constant of motion itself. The first two properties are corollaries of conservation; the latter two give the retention bounds under noise and the thermodynamic cost.

First, no refresh cycle. By the commutation relations (31) and the decoupling construction, the evolution generator $\mathcal{G}$ of a decoupled sector contains no $\mu^{(*)}$ terms of that sector, so $\mathcal{G}^*(N_p) = 0$ and $\frac{d}{dt}\langle N_p \rangle = 0$: the stored content is a constant of motion, and the leak–refresh cycle does not exist. Place-value storage needs refresh because charge is not a conserved quantity of the circuit dynamics.

Second, nondestructive reading. The spectral projections P_m of N_p commute with H , U_χ , and the decoupled generator; the post-measurement state $P_m \rho P_m / \text{tr}$ continues under the same evolution, and the probability of reading m again is one.

Third, retention time. Under the localized symmetric noise model delimited in §4.1.1 and $\beta > 1$, (32) gives $\tau_k = 1/\text{Re } D_k(\beta)$; it is a specification in force once conditional-kernel acceptance has passed, and it does not constitute a retention law for arbitrary baths. The slowest mode of the superselection labels is fixed by the two-sided bounds of (67) at $\Theta(q^{-K})\kappa$; actual device rates must still be measured independently.

Fourth, erasure cost. Heating randomizes an ordered record and is an entropy-increasing process; logical erasure is resetting an unknown logical distribution to a standard state. The environmental heat-absorption lower bound for iso-energetic degenerate memory is (34), a uniform bit giving $k_B T \log 2$; nondegenerate processes use the free-energy-difference bound. The irreversible cost of a complete cycle hangs on the reset/rewrite step, not on the entropy-increasing thermalization step [7, 19].

6.5 Operational Properties

Operations run on two native channels: multiplication acts componentwise on the occupation lattice, and addition is assembled through F in the phase sector; the integrality of the occupation spectrum and the unique factorization of the frequencies jointly determine the precision profile. The first two properties follow from this; the third is given by symmetry-protected readout.

First, multiplication, division, and powers are exact. The action of μ_q on occupation vectors is integer vector addition: $v(qn) = v(n) + e_q$ holds componentwise. Rounding would require occupations to take non-integer values, but the occupation spectrum is $\mathbb{Z}_{\geq 0}$, so rounding does not occur. Approximation appears only at the addition interface, with amplitude controlled by (30).

Second, no crosstalk. Any sum-frequency coincidence $\sum_i a_i \log p_i = \sum_j b_j \log q_j$ exponentiates to $\prod p_i^{a_i} = \prod q_j^{b_j}$, and by unique factorization the two sides are the same factorization: the absence of spurious resonance channels is given by the frequency–unique-factorization relation of §8.1.

Third, automatic coherence allocation. By symmetry-protected readout (§4.1.4), measurable operators O are Galois-invariant combinations, whose expectations contain only the diagonal blocks of conserved labels,

$$\langle O \rangle = \sum_{\text{labels } \ell} \text{tr}(\rho_{\ell\ell} O_{\ell\ell}), \quad (57)$$

and coherences $\rho_{\ell\ell'}$ ($\ell \neq \ell'$) between different labels enter no readout. Decoherence between

distinguishable sectors therefore changes no measurable result: its harmlessness follows from the observable algebra; the degrees of freedom needing coherence protection are exactly the interiors of self-conjugate orbits, i.e., where amplitude-level evaluation happens.

6.6 Input–Output Properties

Input and output both complete through the boundary: writing is modewise preparation or transduction, and readout is QND projection with Born statistics; the cost is decided by the semantic location of the data and the dictionary size. Five properties unfold on this basis.

First, load matching. By (28), multiplicative data satisfying the normalizability condition are product states, preparable mode by mode. Equation (29) gives the entanglement of a two-branch superposition on a concrete prime cut; entanglement is nonzero only when both sides distinguish the branches. The preparation lower bound of a general additive load is decided by its full Schmidt spectrum, not by the label “addition” alone.

Second, the resource lower bound of dictionary size. If K prime modes each take occupations $0, \dots, L$, the state-space dimension is at least $(L + 1)^K$; realizing it with m units of local dimension $\leq d$ requires $m \log d \geq K \log(L + 1)$: packing many independent prime modes into a few bounded-dimension units has no free route. If instead frequency addressing is to host all modes $p \leq N$ within one band, the pigeonhole principle gives $\min_{p \neq p'} |\log p - \log p'| \leq (\log N - \log 2)/(\pi(N) - 1)$, and any frequency-selective operation constrained by the energy–time relation takes worst-case time $\Omega(N/\log^2 N)$. A full dictionary therefore has three basic realizations (and their hybrids), each openly priced: independent spatial modes pay $\pi(N)$ addresses, frequency multiplexing pays resolution and time, and compressed encoding pays a p -dependent controller (whereupon μ_p is no longer a free primitive). The machine’s load-matching claims all presuppose this lower bound and claim no way around it.

Third, two semantics of input encoding. Numeric semantics (using the arithmetic value of n): the conversion from place value to occupation representation and integer factorization reduce to each other: given the factorization of n , the occupation vector is read off; given a converter, its output on input n is the complete prime factorization; both directions are polynomial time. String semantics (data as symbol sequences): the Gödel occupation encoding $s = (s_1, \dots, s_k) \mapsto v_{p_i} = s_i$ writes channel by channel directly, at linear cost, without factorization: bit strings and text-like data enter with no factorization cost. Structured data take divisibility encoding: a graph’s edge set maps to the squarefree number $\prod_{(i,j) \in E} p_{ij}$, subgraph relations are divisibility relations, and graph queries land in the gcd/lcm operations of the divisibility lattice (§6.3). Factorization cost hangs only on numeric semantics.

Fourth, the precision scaling law. The variance of Born-readout estimators contracts as M^{-1} in sample count, $\varepsilon_{\text{stat}} \propto M^{-1/2}$; on the thermodynamic side, the deviation of observables in the state parameters is controlled by response coefficients, $\varepsilon_{\text{th}} \propto |\beta - \beta^*|$. Total precision $\varepsilon = O(|\beta - \beta^*|) + O(M^{-1/2})$ is tunable along two continuous parameters, without word-length jumps.

Fifth, output definiteness. The occupation spectrum is $\mathbb{Z}_{\geq 0}$ and QND readout returns eigenvalues: output is natively discrete, with no analog-style precision blur, and errors are only classification errors (thermal fluctuation misreading a state, at rate $e^{-\beta\Delta}$, §6.9). M independent repetitions with majority vote suppress the decision error exponentially, at logarithmic cost. Verifiable loads (whose answers admit multiplicative-side algebraic checks, as in the register-by-register comparison of §5.1.3) take the Las Vegas protocol: failed verification means rerun, the output is unconditionally correct, and the expected time is finite.

6.7 Control Properties

Control only imposes boundaries (temperature schedules and pump sequences); evolution is executed by the relaxation spectrum. The control properties revolve around three things: a priori computability of schedules, certifiability of convergence, and distinguishability of stagnation.

First, computable timetables. The cooling schedule is jointly determined by target precision, coupling strengths, and lower bounds of the relaxation spectrum (spectral bands in Appendix B, Step 5); its parameters should be recorded before the experiment and rechecked at readout.

Second, convergence conditions. When the specified generator is ergodic, quantum detailed balance holds, and the gap is positive, relative entropy provides a Lyapunov function and fixed-temperature evolution converges to the corresponding Gibbs state. The optimality of (64) applies only within the comparison class of fixed-degree, uniform-edge-weight, reversible regular-graph walks; other geometries and nonlocal or nonreversible dynamics are not constrained by that bound.

Third, a convergence certificate within the conditional kernel. Under the localized symmetric noise model of (32), $\beta > 1$, and estimators that are linear combinations of the corresponding character eigenmodes, the sector- k autocorrelation decays as $e^{-t \operatorname{Re} D_k}$, so for estimators in the linear span of sector observables the integrated autocorrelation time obeys the a priori bound

$$\tau_{\text{int}} \leq \frac{2}{\min_{k \neq 0} \operatorname{Re} D_k(\beta)}, \quad (58)$$

with the right-hand side computed from the explicit series of (32). This bound proves only integrated-autocorrelation control within that eigenmode subspace; total-variation mixing of the full chain still needs conditions such as ergodicity, reversibility, and minimal stationary mass. The correspondence between the sum-of-squares form of positivity arguments and certificate readout appears in §7.

Fourth, quantifiable gaps. If a load's penalty decomposes channelwise as $H_{\text{sep}} = \sum_p h_p(N_p)$, and the bath and Metropolis factors also act channelwise, the generator is the tensor sum $\mathcal{L} = \bigoplus_p \mathcal{L}_p$, whose spectrum is the sum set of the channel spectra (the $\mathcal{L}_p$ act on distinct tensor factors and commute pairwise), so

$$\boxed{\operatorname{gap}(\mathcal{L}) = \min_p \operatorname{gap}(\mathcal{L}_p)} \quad (59)$$

independently of the channel number K ; for warm starts or finite chains with given $\pi_{\min}$, standard bounds [18] still carry initial-state/minimal-stationary-mass terms, e.g., $\operatorname{gap}^{-1}[\log(1/\pi_{\min}) + \log(1/\varepsilon)]$; one may not conclude from the tensor-sum structure alone that K enters only logarithmically. Single-channel gaps follow in closed form from bath rates and penalty strength (for binary chains, the sum of the transition rates), so convergence rates are computable a priori. Numerics: for $K = 2, \dots, 8$ the ratio of full-system generator diagonalization to the $\min_p$ formula is identically 1.000000 (the tensor sum is exact structure; details in Appendix A). Contrast: under cross-channel coupling constraints (number-partition type) the gap collapses exponentially in K (measured $\sim e^{-0.58K}$); this is the glassy spectral form within this finite numerical family. Constraint decomposability gives a sufficient condition for the tensor sum; the hardness of cross-channel constraints still needs family-by-family analysis, and decomposability may not be equated with a general solvability boundary.

Fifth, distinguishing non-halting. When constraints are unsatisfiable, zero-penalty configurations do not exist; under glassy stagnation the target candidate is slow to appear, accompanied

by growing relaxation times and the aging scaling of response functions. The observable difference of the two signatures (occupation magnitude versus relaxation scaling) provides statistical grounds for the distinction; the grounds accumulate evidence and do not constitute a decision (§4.1.4).

6.8 Native Limits

The machine's limits and its capabilities come from the same structure. The five items below are theorem- or axiom-level statements, not engineering limitations.

- (1) The coherent layer is not gate-universal. The phase classes and F generate the Weil representation [16] of the finite group $\text{SL}_2(\mathbb{Z}/q^K\mathbb{Z})$ (end of Appendix B); a finite group is not dense in the unitary group, so the coherent operation set is structurally not a universal gate set; coherence protection is allocated per symmetry sector (§6.5).
- (2) Unmeasurable operator classes exist. Symmetry-protected readout makes self-adjoint combinations of the class $i(a_\chi - a_{\bar{\chi}})$ nonexistent at the measurement level (§4.1.4); the measurable algebra is drawn by Galois invariance.
- (3) The cloning boundary. Conserved charges, once QND-read, can be re-prepared from the record and are copyable information; amplitudes inside self-conjugate orbits inherit the quantum no-cloning theorem. The boundary between copyable and uncopyable is precisely the measurable-algebra boundary of the previous item.
- (4) No stored program. Programs live on the boundary, not in the state space; a classical-style universal machine does not exist; self-reference must close through measurement and return coupling, billed for resources (§3.11).
- (5) No encoding-equivalence axiom. Conversion between representations is a billed physical process whose cost enters the resource vector (25); the divide between numeric and string semantics (§6.6) arises from this.

6.9 Error Model

Errors fall into three classes by origin, each controlled by one gate, with composable budgets; the three are the aggregate form, at the readout end of low-temperature loads, of the seven error classes along the run chain of §3.10.

First, statistical fluctuation. The shot noise of Born readout, controlled by sample count: $\varepsilon_{\text{stat}} = O(M^{-1/2})$ (§6.6).

Second, thermal fluctuation. At finite β non-target configurations carry Gibbs weight; the probability that a low-temperature candidate lands in a non-target configuration is controlled jointly by the degeneracy and $e^{-\beta\Delta}$, the probability fixed by temperature and spectrum together, where Δ is the penalty gap between the target minimum and the next level (the finite configuration spectrum of (46)). If the pinning drive keeps the penalty's zero space invariant, it changes only the gap; general calibration drift can also move the zero space and must be booked as structural error.

Third, structural noise. Galois invariance only restricts the directly readable operator set and is not a general error-correcting code. If the noise is generated by common-mode group actions on conjugate pairs, the σ -fixed subalgebra forms a decoherence-free observable algebra; but

differential-mode phases, sector crosstalk, and general local noise can change the expectations of invariant operators. For example, $a_\chi \mapsto e^{i\delta} a_\chi$, $a_{\bar{\chi}} \mapsto e^{-i\delta} a_{\bar{\chi}}$ changes $\langle a_\chi + a_{\bar{\chi}} \rangle$. That non-invariant combinations lack direct readout channels therefore does not mean noise cannot couple back into measurable components. Code distance, thresholds, and correctable error classes must still be established independently.

When structural noise is absent or its contribution is bounded by calibration as ε_{str} , the total budget reads

$$\varepsilon \leq O(e^{-\beta\Delta}) + O(M^{-1/2}) + \varepsilon_{\text{str}}.$$

The three terms correspond to thermal occupation, statistical sampling, and hardware/control error; only the first two shrink directly with temperature and sample count.

6.10 System-Level Properties and Limits

System-level properties are inherited through the composition semantics and the resource quantifiers: independent composition gives composability, and uniform machine families give universality with its resource ceilings.

First, composability. The statistical operator of independent modules is the tensor product, $Z_{12} = Z_1 Z_2$, $\log Z_{12} = \log Z_1 + \log Z_2$, with energy and cost additive (§5.3.2).

Second, universality and resource ceilings. DPRM gives the unbounded constraint language’s expressive power over recursively enumerable sets; the imperative mode simulates finite-tape Turing machines given an explicit head, a finite controller, and addressing resources. The unbounded universality of both belongs to uniform machine families whose resources grow with the input (§3.11). If mode count, occupation truncation, running time, and inverse error are all polynomially bounded in the input length, and each unitary or open-system segment can be approximated by polynomial-size quantum circuits, the output distributions of this restricted machine family are simulable in the BQP model. Without these resource quantifiers, exponentially small gaps and exponential cooling times cannot be placed inside BQP. This paper claims no complexity-class separation.

Comparison with the prevailing paradigms:

	Place-value machine	Quantum-circuit machine	This machine
Native structure	Addition	Unitary group	Addition $\times$ multiplication
Precision model	Word length	Gate fidelity	Temperature $\times$ samples
Storage model	Refresh cycle	Error-correcting codes	Conserved charges (no refresh)
Coherence strategy	—	Global protection	Allocated per symmetry sector
Error strategy	Detection and correction	Error-correcting codes	Symmetry-protected readout
Convergence condition	—	—	H-theorem and autocorrelation-time bound (58) in the conditional thermal kernel

The engineering realization of these properties appears in §8.

7 Certificate Readout

The readout port of the arithmetic workspace can deliver, along with numbers, finitely many independently verifiable instances of algebraic, spectral, or positivity relations. When the readout target is an arithmetic proposition, stable states, correlation functions, and KMS relations compress into finite positivity certificates that can be rechecked independently: a proof is a non-negative linear combination of finite positivity instances $\varphi(a^*a) \geq 0$ with the operation-algebra identities ((3)–(4)) and the KMS relations.

This semantics organizes into a **certificate system**, made of four pieces: models are non-negative valuations on the object domain (the sieve system takes finitely supported nonnegative weights); axioms are the data or algebraic relations the models must satisfy (the sieve system’s Type I data A_d ; the modular-flow system’s K1 algebraic relations and K2 state conditions); certificates are finite nonnegative combination objects (the sieve system’s pointwise-domination vectors; the modular-flow system takes finite instances of the K2 positivity schema in noncommutative *-SOS combinations); provable bounds are the conclusions certificates yield uniformly over all models. The completeness of the sieve system is given by linear-programming strong duality: provable is equivalent to having no countermodel, and independence proofs are countermodel constructions.

Within this framework the paper establishes five results, with all derivations in Appendix C.

First, the sieve certificate system is complete and admits a parity countermodel: an explicit pair of models makes the positive lower bound for the prime count unprovable in the Type I system, with the upper bound off by at least a factor of 2 (§C.1).

Second, the noncommutative certificate system containing the modular flow σ_t subsumes all reasoning of the sieve system, while the parity witness does not lift (§C.2).

Third, a single $(1, 2, 1)$ positivity instance yields the Mertens inequality; $\zeta(1 + it) \neq 0$ is certified within the system, and the prime number theorem with its arithmetic-progression version follows through the common analytic shell (Tauberian theorems); the two systems separate strictly, and adding the modular flow is a verified strict strengthening (§C.3).

Fourth, sieve asymptotics enter the critical scaling of $\beta \rightarrow 1^+$, with $G(s) = s e^{\gamma + E_1(s)}$ and the Dickman duality $G\hat{\rho} = e^\gamma$ (§C.4).

Fifth, the territory of provability splits into two levels, certification completed and candidate roadmap, with the gap ledger itemized; Siegel zeros are the structural obstruction to explicit constants (§C.5).

This readout semantics places thermalization, phase evolution, and stopping criteria on one computational chain: finite low-temperature readout supplies candidate sectors, and spectra with correlation functions supply verification data; the correctness of a certificate can be rechecked independently of the machine, separating the burden of verification from the cost of running. An engineering realization must therefore preserve modular-flow, phase, and spectral readout; we now turn to the physical conditions for realizing these relations.

8 Engineering

Engineering follows the forward order from relations to devices: every constitutive relation the machine claims lands as a measurable frequency, coupling, thermalization, or readout condition; devices are defined by the relations they carry; acceptance numbers are pre-registered by the numerical twin; and whether a physical platform delivers is settled by the decisive experiments.

8.1 Physical Realization Conditions

A physical system realizing a BC state-space computer must keep the BC covariance relations, constrained thermalization, and observable readout simultaneously in measurable physical degrees of freedom. The following conditions write these relations as experimental requirements on frequencies, couplings, thermalization, and coherence allocation, and give the device landing points of the components of the resource vector (25).

First, frequencies and unique factorization. Take mode frequencies $\{\log p\}$; any sum-frequency coincidence $\sum_i a_i \log p_i = \sum_j b_j \log q_j$ implies $\prod p_i^{a_i} = \prod q_j^{b_j}$, and unique factorization identifies the prime occupations on the two sides. Frequency resolution and linewidth jointly determine the nearly degenerate channels resolvable in experiment.

Second, the thermalization timetable. The cooling law is fixed jointly by the measured relaxation spectrum and the target error (§4.1.3); under the declared Hecke geometry, (64) provides the reference gap.

Third, the channelwise coupling spectrum. The generic matrix element of the addition–multiplication interface is $(q - 1)^{-1/2}$ (30), with large- q scaling $q^{-1/2}$; its calibration is channel-dependent and is recorded together with the truncation error of the finite hierarchy.

Fourth, coherence allocation. The coherence time of self-conjugate-orbit sectors is set by the resolution requirement of scattering readout (§4.1.4); the dissipation of the remaining sectors enters the declared thermalization model.

Fifth, dual-domain encoding. On one carrier, the frequency domain hosts multiplication and the phase domain hosts addition (§8.2); the noise budgets of the two domains are given by frequency stability and phase-decision error respectively.

A finite realization must report its mode set, local truncations, auxiliary degrees of freedom, preparation and reset costs, coherence and dissipation timescales, readout resolution, and the relation residuals of (3)–(4); these entries give the device-side readings of the resource vector (25). Whether the declared constrained evolution is delivered by the device’s geometry, gap, temperature, and energy exchange is judged by the acceptance scheme of §8.4.

8.2 Native Devices

Realization defines devices forward from the machine’s constitutive relations: native means the device’s own physics carries the relation, and zero protocol emulation at minimal resources is the selection criterion; platforms are ranked by how many devices they provide natively, and paths are ordered by verdict power. The devices below belong to the arithmetic module and the thermalization engine; the interaction module’s transduction and actuation are mature industry, with part definitions in §4.2. Prime modes take an oscillator comb with frequencies configured at $\{\log p\}$ (the first 25 primes span a factor of 6.6 in frequency), and seven devices are defined on it:

Device	Relation carried	Native form
Occupation register	N_p , additive energy	Equally spaced level tower. Two cases by the origin of transition-matrix-element flatness. Structural flatness: a tilted lattice for a tagged particle (Wannier–Stark ladder [23]), occupation is lattice position, equal spacing from the tilt, flatness from translation invariance; controlled flatness: an anharmonic tower calibrated level by level, with $\sqrt{n}$ matrix-element growth to be compensated. The origin of flatness is precisely the verdict quantity between the two cases
Multiplication gate μ_q	Occupation translation	Energy-conserving two-body exchange $ j\rangle 0\rangle_W \rightarrow j+1\rangle 1\rangle_W$ (a work mode at gap $\log q$ carries the conservation), parametrically switched; adiabatic ladder climbing as the calibration-robust variant [26]; bare three-wave mixing unusable: μ_q is not the bosonic $a^\dagger$ (off by the $\sqrt{N_q + 1}$ normalization), and occupation encoding has no composite mode ω_{pq}
Thermalization engine	$\mathcal{L}_\beta$, KMS ratio	Two native families. Collisional reservoir: a stream of fresh two-level systems (gap $\log q$, temperature β), the ratio $q^{-\beta}$ given automatically by energy conservation and bath occupation (Davies limit in Appendix B); passive filtered thermal line: a multi-band-pass spectral density windowed at each $\log q$, fully passive and always on. Neither family sets rates one by one; the degeneracy (character) register is the only device yet to be invented
Addition and F	V_c, F_q, C	The modular structure of the phase ring is given by topology; F takes two passive cases, a q -port DFT interference network or a dispersive chirp transform, both structure rather than control; the quadratic phase C is a diagonal phase mask, same mechanism as V_c , completing the SL_2 closure ((60))
QND probe	Charge and sector readout	Dispersive-coupling measurement, the Stark shift χN compatible with ladder linearity; target-sector occupation detected by power or QND detection
Program layer	Boundary b	Pump spectra, selection rules (channels pre-detuned, pump compensation switches them on), and parametric pinning; the red-line criterion pair (the non-negotiable criteria separating physical thermalization from protocol emulation) is the control-bandwidth audit and the counterfactual physical scan
Interaction front end	$\mathcal{T}_A, \mathcal{W}_b$	Transducers map physical features to $(\Delta N, \Delta\varphi, E_i)$, actuators the reverse; QND conserved charges trigger pump switching, and legacy controllers serve only as first-machine stand-ins and compatibility boundaries

The physical form of the addition unit carries two encoding slots: the frequency slot of a

carrier $e^{i(\omega t + \varphi)}$ is the multiplication register, where mixing is frequency addition is numerical multiplication; the phase slot is the addition register, where phase shifting is phase addition is numerical addition, with modular reduction given natively by the ring structure of phase; the physical landing of F is the two passive cases of the table above. The mod- q addition unit consists of three pieces:

(1) Encoding and phase shifting: residues are encoded as phases $\varphi_a = 2\pi a/q$, and cascaded phase shifters realize $\varphi_a + \varphi_c$ (the operator identity is the phase class V_c , (6)); the ideal operator relations are exact, and device error is given by phase stability.

(2) Decision readout: interference against a reference arm makes the q -point phase decision (phase-shift-keying demodulation; noise within the threshold is removed by decision regeneration).

(3) Coprime parallelism and zero carries: coprime modulus channels run in parallel with zero carries between them: the delay main term of place-value adders (carry propagation) has no counterpart; when the modulus channels are addressable in parallel, the single-round modular-addition depth is $O(1)$, with channel count, addressing, and final CRT reconstruction charged separately. Residue-to-place-value reconstruction happens only at the output boundary; the machine's internal multiplication, comparison (H readout, §6.2), and pinning consume the residue representation directly.

The precision budget splits between the two domains. First, multiplicative-side frequency stability: the gap floor $\log(1 + 1/n_{\max})$ set by adjacent combination frequencies ($n_{\max}$ the largest integer in the energy window) provides the reference for the oscillators' relative stability; the floor narrows inversely with $n_{\max}$, so the linewidth requirement of the coherent multiplication layer tightens exponentially with the energy window; this item is booked per §3.9, and relaxation-tier loads are not constrained by it. Second, additive-side phase stability: the q -point decision requires phase noise below π/q , giving the single-channel modulus ceiling. Frequency drift harms multiplication, phase jitter harms addition, and the two budgets are mutually independent.

Scaling takes a partitioned architecture. The cost of growing the channel count is quantified first: the comb-tooth spacing, by the average prime gap, is $\log(p_{k+1}/p_k) \sim \log p_k/p_k$ ($\sim 10^{-3}$ at $K=1000$), so stability requirements narrow polynomially; the bottleneck lies in pump-multiplexing intermodulation (channel pairs K^2) and readout multiplexing. Channel locality (single-channel action of the multiplication gates, decomposable constraints, and the crosstalk bounds given by frequency resolution) keeps inter-cluster coupling sparse, so take a c -cluster partition: each cluster hosts K/c channels with an independent frequency band and mixing bus, and only readout aggregation and $O(c)$ addition-interface channels cross clusters; the gap is unchanged (the tensor sum of (59) holds across clusters), frequency and pump budgets are per cluster, so decomposable-constraint loads scale linearly with no global bottleneck; amplitude-level loads need cross-cluster coherence and are outside this clause. The occupation-truncation error has an a priori closed form: Gibbs weights decay geometrically, truncation at ℓ levels errs $\sim p^{-\beta\ell}$, and the criterion for sensitive loads is their maximal occupation demand.

8.3 Decisive Experiments

The decisive experiments can be planned at single-cavity level and sit at tier zero of the platform tiers (§8.5); all verdict quantities are parameter-free, with the numbers issued in advance by the numerical twin (§8.4).

First, the flatness duel. The two occupation-register cases duel on ladder-climbing rates: the harmonic oscillator gives $\sqrt{j+1}$ growth, the tower gives a flat line; the signature is parameter-

free, and the winner fixes the carrier.

Second, the red-line verdict. The winning tower plus the thermalization engine: the detailed-balance ratio $1/7$, the $7:1$ mirror under reversed pairing direction, and the tower-depth sequence $0.4041 \rightarrow 0.3405$ tending to 0.3386 .

Third, the criterion pair. Non-protocol-emulation is verified jointly by two criteria: the counterfactual physical scan (change the bath temperature or the gap without touching the controller; the ratio must track $q^{-\beta}$ automatically), and the control-bandwidth audit (the information rate of the control channel is insufficient to specify the trajectory).

Fourth, the character-register verdict. The verdict quantities are parameter-free ratios of the three timescales: radial-band center to character relaxation $(q+1):1$, and radial gap to character relaxation $(\sqrt{q} - 1)^2:1$; measuring the ratios is measuring the tree geometry.

The additive side's passive F verification (winding closure, q -point decision error rate, covariant interference $V_c F|a\rangle = F|a+c\rangle$) is a desktop side line and can go first.

8.4 Numerical-Twin Acceptance

Acceptance unfolds along four levels: single-channel thermalization, gate set and group closure, tower and three timescales, and multi-prime load. The four levels are booked by device: stage one tests the occupation register and the thermalization engine (including the red-line criterion pair), stage two the multiplication gate and the addition unit, stage three the character register (the yet-to-be-invented device beyond the seven, its verdict quantities the two parameter-free three-timescale ratios, §8.3, Fourth), and stage four program-layer reuse and end-to-end load. The numerical twin solves the dynamics under the specified KMS collisions, Hecke geometry, and boundary conditions, checking verdict quantities, analysis chains, and acceptance software, yielding sixteen verdicts and three implementation corrections.

Physical-platform testing is kept separate from the numerical twin. Experiments take temperature, gap, equal-coupling geometry, and one overall timescale as inputs, reconstruct the generator from multi-initial-state $P_j(t)$, and compare against the pre-given ratios, gaps, and degeneracy relations. Controllers that set the $1:q$ jump ratio or the full spectrum by hand are protocol re-enactments, not physical tests of the thermal kernel; non-Hecke geometries at the same temperature and gap provide the corresponding controls.

Stage one: single-channel thermalization. Protocol: an n -level equally spaced ladder (gap $\log q$) and a single two-level thermal mode (KMS occupation $q^{-\beta}/(1 + q^{-\beta})$) in repeated interaction (collision model [24]), the collision unitary being the energy-conserving exchange $U = \exp[-i\theta(V \otimes \sigma^- + V^\dagger \otimes \sigma^+)]$, with a fresh bath injected after each collision. Verdicts and results ($q = 7$): First, the flatness duel. The structurally flat case's level-by-level climbing-rate coefficient of variation is zero within numerical precision, while the harmonic control's level-by-level rates fall on the $\sqrt{j+1}$ law (fit residual 4.4×10^{-5}): the carrier-selection verdict is parameter-free (§8.3, First).

Second, the detailed-balance verdict. The steady state deviates from the truncated Gibbs distribution by at most 6.2×10^{-16} , with shell ratios $1/7$ to eight digits; with the pairing direction written backwards the steady state mirrors to $7 : 1$, and this verdict also checks the direction of energy conservation.

Third, the counterfactual scan. With the controller parameters frozen throughout ($\theta = 0.35$), changing the gap to $\log\{5, 7, 11\}$ ($T = 1$) gives ratios $0.200000/0.142857/0.090909$, and changing the temperature to $T = 5/4$ and $2/3$ ($q = 7$) gives $0.210825/0.053995$, tracking the truncated Gibbs pointwise with deviation $\leq 1.9 \times 10^{-13}$; the control channel carries only θ and the switching

times, an information rate insufficient to specify the trajectory: the twin form of the red-line criterion pair (§8.3, Third).

Fourth, two-family equivalence. The collisional reservoir and the passive filtered thermal line (a Lindblad birth–death chain) share the Gibbs fixed point (deviation $\leq 3 \times 10^{-15}$); after normalization by $\sin^2 \theta(1 - p_e)$ the collision gap equals the thermal-line gap (deviation $\leq 5.5 \times 10^{-14}$ at all of $\theta = 0.35, 0.15, 0.06$): hardware may choose either family, and the acceptance numbers are common.

Fifth, tower-depth convergence. The classical-chain gap at layer counts $n = 7, 15, 25, 41$ is 0.4041, 0.3530, 0.3438, 0.3405, monotonically approaching the asymptotic value 0.33856 of (64).

Sixth, monotone decay of inter-shell coherence. Shallow-tower verdict values must be given per layer count: the $n = 7$ platform’s verdict value is 0.4041 (the asymptotic 0.3386 belongs to deep towers).

Stage two: gate set and group closure. The gate set is the cyclic shift X , the scaling permutations M_g , the quadratic phase $C = \text{diag}(e^{2\pi i \bar{2}x^2/q})$ ($\bar{2} = 2^{-1} \bmod q$), and the interface F . First, group closure. Phase-invariant exhaustive closure gives the closure tower ($q = 7$):

$$\langle X, M \rangle = 42 = |\text{Aff}(\mathbb{F}_7)|, \quad \langle C, F \rangle = 336 = |\text{SL}_2(\mathbb{F}_7)|, \quad \langle X, M, F, C \rangle = 16464 = 49 \times 336, \quad (60)$$

the last entry being the order of the semidirect product of the finite Heisenberg group (mod center) with $\text{SL}_2(\mathbb{F}_7)$ (the Jacobi group): the gate set contains the translation–modulation layer together with its automorphisms. Removing C , $\langle X, M, F \rangle = 588 = 49 \times 12$, keeping only the semidirect product of the Heisenberg layer with the torus normalizer, so the quadratic phase gate is the independent operation this closure requires. Second, relation and duality verdicts. The relation verdicts are $F^2 = \text{parity}$, $F^4 = 1$, the Bruhat cell count $42 + 42 \times 7 = 336$, and the F -eigenvalue multiplicities $(2, 2, 2, 1)$, all with residuals below 10^{-14} ; the quadratic-residue character is an eigenvector of F with eigenvalue i (the Gauss sign for $q \equiv 3 \bmod 4$, residual 1.3×10^{-15}). The duality verdicts: FXF^{-1} is a phase gate and $FM_gF^{-1} = M_{g^{-1}}$ (residual 4×10^{-15}). Under unitary perturbations of strength ε on F, C , the relation residuals are linear over three decades (slope 0.90); a residual cap of 3% corresponds to a mean gate fidelity of about 99.97%.

Stage three: tower and three timescales.

First, the dispersion verdict. The eigenpairs (θ_m, μ_m) of the $n = 41$ symmetrized spectrum fall on the curve $\mu = 1 - \frac{2\sqrt{7}}{8} \cos \theta$ with deviations 0.02%–0.57%; the measured relaxation-rate band $[0.3405, 1.6595]$ against the asymptotic $[0.3386, 1.6614]$.

Second, ballistic drift. The mean layer index $\langle j \rangle(t)$ of a tower-top initial state has slope -0.75013 against the analytic value $-(d - u) = -0.75$, with linear residual 0.003.

Third, primitive-character degeneracy. All thirty-six primitive characters mod 49 decay at rate 0.125000, with maximal deviation 8.2×10^{-16} : the channel degeneracy of (66) hits channel by channel.

Fourth, superselection scaling. The slowest nonzero modes of the full-ring generator at $K = 2, 3, 4$ are 0.014065, 0.001901, 0.000268, adjacent ratios 0.135, 0.141 tending monotonically to $1/7$, consistent with the two-sided bounds and asymptotic constant of (67). Simulation-protocol note: when stationary-measure ratios reach 7^{40} , non-symmetric eigendecomposition is numerically ill-conditioned; time-domain quantities always use the symmetrized spectrum or direct integration.

Stage four: multi-prime, tree geometry, and native load.

First, two-prime tensor thermalization. v_2 (six levels) and v_7 (four levels), each with its own two-level KMS bath at the same temperature $\beta = 1$ and frequencies $\log 2$ and $\log 7$, collide alternately to steady state: the joint distribution deviates from $\text{Gibbs}_2 \otimes \text{Gibbs}_7$ by at most 1.6×10^{-15} , the ratios $1/2$ and $1/7$ hold simultaneously, and the mutual information is 5.7×10^{-17} , consistent with the Euler factorization of $H = \sum_p (\log p) N_p$.

Second, radial lumpability of the tree geometry. On a depth-four sevenfold horoball tree (2801 vertices), a single-leaf initial state evolves for sixty steps; the radial marginal deviates from the one-dimensional chain by at most 1.2×10^{-13} , while the top-layer branch occupations are $[0.988, 0.002 \times 6]$: branch labels are limited by the tower-top bottleneck.

Third, the scattering-evaluation load. At truncation $N = 2 \times 10^5$, $\sigma = 0.8$, the Gibbs-probe autocorrelation deviates from $\zeta(2\sigma - it)/\zeta(2\sigma)$ by $6.5 \times 10^{-5} - 2 \times 10^{-4}$, of the same order as the truncation tail bound; $L(2\sigma - it, \chi \bmod 7)$ is reproduced at the same precision by the one-sided U_χ cross-correlation protocol of §5.1.2: the end-to-end acceptance of (45).

Fourth, boundary evidence from the annealing load. Factorization recovery of the target $N = 392 = 2^3 \cdot 7^2$, with logarithmic penalty $E = (\sum_p v_p \log p - \log N)^2$, is a cross-channel coupling constraint: the penalty gap $\Delta = 0.0178$ is computable in closed form, but the nearly degenerate configurations $(6, 1)$, $(0, 3)$ and the inter-valley barrier quench single-step proposals at final temperature $\beta = 900 \gg 1/\Delta$ (success rate 0.30–0.38). Per the Las Vegas protocol of §6.6, with multiplicative-side algebraic checks and restarts, the output is delivered after verification, at 3.4 attempts on average.

Tolerance windows are pre-registered together with the verdict quantities (scanned by the same twin code family): the flatness verdict is safe under level-coupling disorder $\sigma \lesssim 5\%$; the ratio is immune to bath detuning (up to detuning equal to the collision strength) and to 30% per-collision strength jitter, the thermalization engine’s two calibration-free properties; parasitic zero-temperature loss $g \lesssim 10^{-3}/\text{collision}$ keeps 1% ratio precision (the analytic model $\text{err} \approx g/(\sin^2 \theta p_g)$ agrees with numerics to two digits); the character register holds ν within a 1% band under 10% rate disorder, while 1% missing edges shift it by -1.6% , making fabrication completeness its primary process metric.

In the specified numerical condition model, the sixteen verdicts agree with the pre-given relations. The shallow-tower gap, the quadratic phase gate, the restart protocol for glassy loads, and the tolerance windows above thereby enter the hardware acceptance specification; whether a physical platform yields the same set of relations remains for independent experiments to decide.

8.5 Platform Tiers

Hosts are tiered by how many devices they provide natively; coherence demands are set by the coherence allocation (§6.5).

Tier one, classical annealing networks: numerical oscillator banks (FPGA or analog circuits) plus digital noise baths, room temperature, tens of channels, validating protocols and emergent quantities; their bath is set rate by rate by a controller and structurally cannot judge the red line.

Tier two, optical parametric oscillator networks: a retrofit of existing Ising-type annealers [9], with the frequency configuration changed to $\{\log p\}$ and the spin coupling matrix replaced by conservation matching and the penalty terms (46); subset-product loads occupy binary values, one degenerate parametric oscillator per prime; measurable are heat exchange, finite-size low-temperature concentration, and the penalty floor; the retention condition is all-optical coupling; measurement-feedback variants realize the coupling matrix by an in-loop controller and belong

to protocol emulation.

Tier three, superconducting parametric cavity arrays [21]: the self-conjugate-sector coherence protection and the dispersive QND readout required by scattering evaluation are realized here, the full-coverage candidate for all seven devices.

Two backup platforms: trapped-ion Fock-state control [22] is suited to the single-tower verdict; synthetic frequency dimensions can construct coupling-graph geometries freely, the candidate hunting ground for the character register and the tree geometry. Concrete noise effects are checked by the error model of §6.9 and the run-chain classification of §3.10.

8.6 Benchmarks

The benchmark is restricted to the arithmetic workspace, with value range $[2, 10^6]$ and 3×10^5 steps (method details in Appendix A); it tests the mixing and readout of the specified samplers within that workspace and does not define the performance of the whole state-space machine.

Multiplicative target $\pi \propto 3^{\Omega(n)} n^{-1.2}$, with exactly enumerable ground truth ($\mathbb{E}[\Omega] = 13.8399$). Arithmetic annealer: estimate 14.01, integrated autocorrelation time 554, acceptance rate 0.33. Place-value random-walk Metropolis baseline (four tuned window widths): all fail to converge, estimates 6.6–15.3, autocorrelation times 6×10^3 – 1.9×10^4 . The mechanism is (47) run in reverse: adjacent integers have independent multiplicative structure, and additive proposals from a high-weight state land almost surely in low-weight states.

Scaling separation. The same multiplicative target scanned along value-range size N (2×10^5 steps, baseline tuned over four window widths per N , best taken):

N	Truth $\mathbb{E}[\Omega]$	Annealer est/IAT	Best baseline est/IAT	Separation
10^4	8.402	8.543 / 318	8.299 / 1030	3.2
10^5	11.093	10.928 / 547	10.609 / 12800	23
10^6	13.840	13.947 / 486	15.376 (not converged) / 18094	∞

The annealer’s autocorrelation time is flat in N (318/547/486 across three decades), with all estimates converged; the baseline grows with N until failure (at $N = 10^6$ the estimate misses truth by 11%). The mixing-time scaling exponents of the two samplers are of different classes: a single-point qualitative gap rises to a scaling separation. The flatness is consistent with (59): the gap of the divisor-lattice walk is decided by channel structure, independent of value-range size; the baseline’s failure curve is the measured form of the addition–multiplicativity mismatch deepening with scale.

Additive target $\pi \propto \exp(-|n - n_0|/\sigma)$ (narrow peak). The arithmetic annealer’s acceptance rate is 0.000: failure; the baseline’s autocorrelation time is 13: success. The two-way outcome jointly gives load matching: the algebraic structure of the proposal kernel must match the algebraic structure of the target.

Mixed-kernel experiment: with the additive mixing kernel taking the spectral shape $\Lambda(q)/q$ (Mertens-type heavy tail, no hyperparameter), the narrow-peak target’s autocorrelation time is 20, close to the tuned baseline without needing target-scale information. The mixing ratio of the two kernels has no universal value: the ratio is a context parameter, the shape a structural parameter.

Scope statement for the benchmark conclusions. The separations above are matching effects between proposal kernels and target algebraic structure, and constitute no advantage claim in the sense of algorithmic complexity: on the finite subset-product sector, prime-factor proposals

and coordinate-flip Metropolis on exponent vectors are stepwise isomorphic under the same random stream, and any classical implementation knowing the factorization can replicate the annealer orbit by orbit. The correct reading of the benchmark is load matching: matching the proposal coordinates to the target algebra can markedly improve finite-instance autocorrelation times. Within the class of fixed-degree, uniform-edge-weight, reversible regular-graph walks, the Hecke geometry attains the Ramanujan bound; this in-class result does not exclude that other geometries or nonlocal dynamics are faster. The benchmark constitutes no asymptotic complexity advantage.

8.7 Scope and Limits

Under the resource stipulation that mode count, occupation truncation, running time, and inverse error are all polynomial in the input length, and that each channel segment is approximable by polynomial quantum circuits, this machine family is simulable by the BQP model [33, 34]; without that stipulation no complexity-class containment is claimed. The numeric-semantics conversion from place-value input to occupation output is inter-reducible with integer factorization and is constrained by the dictionary-size lower bound (§6.6). The Type I sieve interface carries the parity barrier, but this barrier is not a uniform information-theoretic ceiling for all negative loads (Appendix C.3).

Advantage claims are limited to two kinds: load matching in finite benchmarks, and Ramanujan optimality within the fixed-degree, uniform-edge-weight, reversible regular-graph class. KMS locks the detailed-balance ratio but neither the total rate nor general relaxation kernels; superselection timescales are fixed by the two-sided gap bounds of (67). This paper claims no asymptotic acceleration of arbitrary external bit-string algorithms. Applicable loads include multiplicative-structure sampling, finite exact rational arithmetic, analytic-function response evaluation, and verifiable constraint-satisfaction low-temperature search; performance is jointly decided by input conversion, joint gaps, addressing, and readout costs. Two load classes sit on the limit side: high-speed servo loads whose environmental correlation bandwidth exceeds the tracking condition (53), and strongly coupled and glassy loads (gap criterion in §6.7).

A Benchmark Details

Sampler implementation: first 25 primes, proposal weights $\Lambda(p)p^{-3/2}$, up and down each half, symmetric across edges so the acceptance is the smaller of the target ratio and 1. Baseline window widths $W \in \{10, 10^2, 10^3, 10^4\}$. Autocorrelation times estimated by the window method (truncation threshold 0.05). Ground truth enumerated by a smallest-prime-factor sieve computing Ω up to 10^6 . The mixing experiment scans the ratio $f_{\text{add}} \in \{0, 1/49, 1/7, 1/2, 6/7, 1\}$, cross-testing both target types.

Scaling-separation scan (§8.6): 2×10^5 steps, burn-in 2×10^4 ; IAT by the window method with threshold 0.05; baseline widths $W \in \{10, 10^2, 10^3, 10^4\}$ tuned independently per N , the one nearest truth taken as best; convergence criterion, estimate within $\leq 5\%$ of truth; truth by full Ω sieve enumeration.

Gap numerics (§6.7): first K primes with binary occupations, Metropolis rates, $\beta = 1.5$, penalty strength 1; $K = 2, \dots, 8$ full-system 2^K -dimensional generators directly diagonalized. Frustration-free (channelwise constraints): the gap-to- $\min_p$ -formula ratio is 1.000000 in all seven cases. Frustrated control ($\sum v_p \log p \approx T$ type): the gap collapses from 0.50 ($K=2$) to 0.013 ($K=8$), fit $\sim e^{-0.58K}$.

B Six-Step Derivation of the Conditional Thermal-Kernel Mechanism

This appendix gives the complete derivations of the six conclusions listed in §4.1.3; the conditions and the register of validity are as declared in that subsection, and the notation follows the main text.

Step 1, KMS locking of the detailed-balance ratio. Raising or lowering one occupation quantum of a single mode costs energy $\log q$, so any jump process compatible with a KMS environment [13] at inverse temperature β must satisfy

$$\frac{\Gamma_{\uparrow}}{\Gamma_{\downarrow}} = e^{-\beta \log q} = q^{-\beta}. \quad (61)$$

The KMS condition locks only this ratio; the total rate κ is not constrained by the equilibrium structure and takes on the clock role (§4.1.4). Below, κ is the rate unit.

Step 2, branching geometry. In the rank-two Hecke candidate geometry above, the vertices of the Bruhat–Tits $(q+1)$ -regular tree [15] are homothety classes of lattices, with adjacency given by the Hecke double coset, whose coset decomposition

$$K \begin{pmatrix} q & 0 \\ 0 & 1 \end{pmatrix} K = \bigsqcup_{b=0}^{q-1} \begin{pmatrix} 1 & b \\ 0 & q \end{pmatrix} K \sqcup \begin{pmatrix} q & 0 \\ 0 & 1 \end{pmatrix} K, \quad (62)$$

shows that each vertex has exactly $q + 1$ neighbors; a tree step is the composite of additive translation and scaling, so the operation set composes existing actions and adds no primitive. Choosing an end and taking the Busemann coordinate j (on the machine's local model j is precisely the q -adic valuation), each vertex has exactly one edge with $j \rightarrow j + 1$ and q edges with $j \rightarrow j - 1$. With equal edge weights,

$$P_{\uparrow} = \frac{1}{q+1}, \quad P_{\downarrow} = \frac{q}{q+1}, \quad \frac{\pi_{j+1}}{\pi_j} = \frac{P_{\uparrow}}{P_{\downarrow}} = \frac{1}{q}, \quad (63)$$

and the radial detailed-balance ratio automatically equals the reading of (61) at the critical point $\beta = 1$: the branching number itself realizes the Boltzmann factor, with no need to tilt edge weights. For general β take the normalized radial kernel $P_{\uparrow} = 1/(1 + q^{\beta})$, $P_{\downarrow} = q^{\beta}/(1 + q^{\beta})$; its gap has the closed form $\text{gap}_{\beta} = 1 - 2q^{\beta/2}/(1 + q^{\beta})$ ($\beta = 0.8, 1, 1.2, 1.5, 2$ give 0.242, 0.339, 0.433, 0.559, 0.720, $q = 7$).

Step 3, the weak-coupling limit. Take the system $H_S = (\log q)N$, a KMS environment mode at frequency $\log q$, and the interaction $H_{\text{int}} = g(V_q \otimes a + V_q^{\dagger} \otimes a^{\dagger})$ (V_q the normalized shell-raising isometry). For a finite device with occupation truncation $N \leq L$, the hypotheses of Davies' theorem [10] hold item by item: the system space is finite-dimensional and H_S bounded; the equal spacing of the occupation ladder makes the secular decomposition of the coupling V_q involve only the Bohr frequencies $\pm \log q$, so the secular average is well defined; $V_q^{(L)}$ is bounded; integrability of the KMS bath correlation function is the only external condition. The weak-coupling limit therefore exists, and the secular generator is the GKSL semigroup [14] with jump ratio (61), whose quantum detailed balance yields relative-entropy monotonicity. A numerical check on the $L=4$ truncated chain gives relative entropy $2.342 \rightarrow 0$ and steady-state residual 3×10^{-18} . Weak-coupling convergence without truncation (H_S unbounded) is a domain condition

that remains open and is not used in this paper: all device-level conclusions are stated at finite truncation.

Step 4, optimality of the spectral gap. The radial chain's gap has the closed form

$$\text{gap} = \frac{(\sqrt{q} - 1)^2}{q + 1} = 1 - \frac{2\sqrt{q}}{q + 1}, \quad (64)$$

where $2\sqrt{q}/(q + 1)$ is exactly the Kesten spectral radius of simple random walk on the $(q+1)$ -regular tree [11]. The Alon–Boppana theorem asserts that the second-largest eigenvalue of any $(q+1)$ -regular finite graph is at least $2\sqrt{q} - o(1)$; graphs attaining the bound are the Ramanujan graphs, whose explicit constructions come precisely from $\text{PGL}_2(\mathbb{Q}_q)$ lattices [12]. Within the comparison class of fixed degree $q + 1$, uniform edge weights, and reversible regular-graph walks, this Hecke geometry therefore attains the Kesten–Alon–Boppana asymptotic bound. Raising the degree, adding long-range edges, or using nonreversible lifts fall outside the class and may achieve different mixing rates; optimality here refers only to the class.

Step 5, the full structure of the relaxation spectrum and the three timescales. The radial eigenfunctions are given by the standard spherical-function theory of the $(q+1)$ -regular tree, classified by a parameter s with eigenvalues $q^s + q^{1-s}$: $s = 1$ gives normalized eigenvalue 1, the equilibrium state; all nontrivial relaxation modes sit on $\text{Re } s = 1/2$, with dispersion

$$\lambda(t) = \frac{2\sqrt{q} \cos(t \log q)}{q + 1}, \quad t \in \left[0, \frac{2\pi}{\log q}\right), \quad (65)$$

and continuous-time relaxation-rate band $[1 - 2\sqrt{q}/(q + 1), 1 + 2\sqrt{q}/(q + 1)] \kappa$. Two slower families of modes exist, composing the three-timescale hierarchy (strictness of the hierarchy requires $q \geq 5$; see (68)).

First, the primitive-character annihilation lemma: descending branches uniformize the fibers, and primitive Dirichlet characters mod q^m sum to zero on every fiber, so the ascending branch is a strict annihilator of primitive-character components. For the $\beta = 1$ equal-weight finite-tower generator defined in this paper, the linear damping of the primitive-character channels is

$$\nu = \frac{\kappa}{q + 1}, \quad (66)$$

degenerate across all nontrivial channels ($q = 7$ numerics: under the full generator on $L^2(\mathbb{Z}/49)$ the primitive components decay at 0.125000, two-time consistent). It sits below the radial band: the damping ratio $\nu/\text{gap} = (\sqrt{q} - 1)^{-2}$ is a zero-parameter dimensionless prediction ($q = 7$: 0.369).

Second, the mixing rate of the superselection labels has provable two-sided bounds for the finite-tower generator of this paper. The full-ring generator is the symmetric Markov chain on $\mathbb{Z}/q^K\mathbb{Z}$ with edge set $\{n, qn \bmod q^K\}$ ($n \neq qn$) and rate $\kappa/(q + 1)$ per edge: ascending edges are the mod- q^K localization of the scaling μ_q , descending edges its adjoint; the chain is reversible with respect to the uniform measure, layer-measure ratios are automatically $1/q$, the full-ring realization of (63). Teichmüller labels are conserved under all raising and lowering on nonzero elements and mix only through $n = 0$; the rate is bounded in four substeps.

Substep 1, sector decomposition. The multiplicative action $R_u f(n) = f(un)$ of the Teichmüller lift $\omega : (\mathbb{Z}/q\mathbb{Z})^\times \rightarrow (\mathbb{Z}/q^K\mathbb{Z})^\times$ commutes with the generator, and $L^2(\mathbb{Z}/q^K\mathbb{Z})$ decomposes into $q - 1$ tame character sectors. Sector functions of nontrivial characters vanish at $n = 0$; moreover $v_q(u - 1) = 0$ for $u \neq 1$, so the action on nonzero elements is free, the sector dimension

is $(q^K - 1)/(q - 1)$, and the sector block is character-independent: all nontrivial sectors are isospectral, with multiplicity $q - 2$.

Substep 2, tree decomposition. In the basis of Teichmüller orbits, the transition graph of a nontrivial sector is a q -ary tree of depth K : the nodes of layer j ($0 \leq j \leq K - 1$) are the wild classes $(j, \bar{u})$, $\bar{u} \in (1 + q\mathbb{Z})/(1 + q^{K-j}\mathbb{Z})$, with an ascending edge truncating one wild digit; the ascending edges of layer $K - 1$ end at $n = 0$, where sector functions vanish, an absorbing edge of rate $\kappa/(q+1)$ at the root, all edges of equal conductance under the uniform measure. Symmetric–antisymmetric splitting over children, layer by layer, reduces the sector spectrum to a union of radial birth–death chains R_j of depth $j = 1, \dots, K$ (layer weights $\propto q^{-j}$, same-rate absorption at the top), with multiplicity $(q - 1)q^{K-1-j}$ for R_j ($j < K$) and R_K the radial part; zero-extension of shallow-chain eigenfunctions gives $\lambda_{\min}(R_{j+1}) \leq \lambda_{\min}(R_j)$, so the superselection gap is exactly $\lambda_{\min}(R_K)$. Numerics at $q = 7$, $K = 3$: the 57-dimensional sector spectrum coincides entry by entry with $R_1^{\times 42} \cup R_2^{\times 6} \cup R_3^{\times 1}$.

Substep 3, the upper bound. Take the sector test function $f(n) = \chi(\omega^{-1}(nq^{-v_q(n)} \bmod q))$ ($n \neq 0$), $f(0) = 0$: the Dirichlet form is supported only on the $q - 1$ edges into 0, and the Rayleigh quotient gives $\text{gap}_{\text{sel}} \leq (q - 1)\kappa/((q + 1)(q^K - 1))$, the precise form of the tower-top conductance estimate.

Substep 4, the lower bound and asymptotics. In rate units of κ , solve for R_K the absorption-time equation $-Lh = 1$ (with $h = 0$ at the absorbing top): the birth–death structure yields the explicit solution

$$h_j = \frac{q + 1}{(q - 1)^2} (q^{K+1} - q^{j+1}) - \frac{q + 1}{q - 1} (K - j),$$

verified by pointwise substitution. Two lemmas pinch the lowest rate: first, the positive lowest eigenfunction satisfies $\varphi = \lambda_{\min}(-L)^{-1}\varphi$, giving at its maximum point $\lambda_{\min} \geq 1/\max_j h_j = 1/h_0$; second, using h as a test function and $\langle h, h \rangle_\pi \geq \langle h, 1 \rangle_\pi^2$ gives $\lambda_{\min} \leq \langle h, 1 \rangle_\pi / \langle h, h \rangle_\pi \leq 1/\mathbb{E}_\pi[h]$. Hence

$$\boxed{\frac{1}{h_0} \leq \frac{\text{gap}_{\text{sel}}}{\kappa} \leq \frac{1}{\mathbb{E}_\pi[h]}, \quad \frac{\text{gap}_{\text{sel}}}{\kappa} = \frac{(q - 1)^2}{(q + 1)q^{K+1}} \left(1 + O(Kq^{-K})\right)} \quad (67)$$

for all odd primes q and $K \geq 2$: both ends expand via h_j to the same $(q - 1)^2(q + 1)^{-1}q^{-K-1}(1 + O(Kq^{-K}))$, so $\text{gap}_{\text{sel}}/\kappa = \Theta(q^{-K})$ with the adjacent- K ratio tending to $1/q$, and the conductance upper bound of Substep 3 is compatible with this. Numerics: at $q = 7$ the slowest nonzero modes for $K = 2, 3, 4$, namely 0.014065, 0.001901, 0.000268 (adjacent ratios 0.135, 0.141), fall inside $[1/h_0, 1/\mathbb{E}_\pi[h]]$, and $q = 11, 13$ with $K \leq 7$ all hit likewise; the two-sided relative width at $q = 7$ narrows as 1.4×10^{-3} , 2.5×10^{-4} , 4.3×10^{-5} for $K = 4, 5, 6$, exponentially at the order Kq^{-K} . The three-timescale hierarchy ($q \geq 5$; the middle inequality is equivalent to $(\sqrt{q} - 1)^2 > 1$) is thereby written:

$$\text{gap}_{\text{sel}}^{(K)} \ll \frac{\kappa}{q + 1} < \left(1 - \frac{2\sqrt{q}}{q + 1}\right) \kappa \leq \gamma_{\text{rad}} \leq \left(1 + \frac{2\sqrt{q}}{q + 1}\right) \kappa. \quad (68)$$

All three ratio families are decided by q in the equal-weight critical model: the first two in closed form, the third bounded by (67) (for $K \geq 2$ the upper bound does not exceed $(q + 1)^{-2}$; the hierarchy is strict for $q \geq 5$). In-band oscillation frequencies are quantized at $\log q$, consistent with this model's radial clock.

Step 6, provenance and validity window of the nonlinear layer. The lowest-order invariant allowed by the channel conservation laws (the selection rules of §5.3.2) is cubic, with structure constants forced to the character convolution $f_{ijk} = \delta_{\chi_i \chi_j, \chi_k}$; vertex amplitudes carry square roots of the couplings (amplitude-level objects). Taking Heisenberg expectation equations of the GKSL dynamics and the mean-field truncation $\langle a_i a_j \rangle = A_i A_j + C_{ij}$, discarding the connected part C_{ij} , gives the corresponding nonlinear resonance equations. In the tested three-mode instance the truncation error exhibits the self-similar law in the combined variable gt , $\text{err}(g/2, 2t) = \text{err}(g, t)$ ($4.29\% \leftrightarrow 4.30\%$, $18.76\% \leftrightarrow 18.91\%$), with a fitted bound of the form $C(gt)^2 e^{\Lambda gt}$, error $< 5\%$ in the test window $gt \lesssim 0.5$. The window has an a priori bound. Suppose the mode moments are bounded on the window $[0, T]$, $\sup_{t \leq T} \langle a_k^\dagger a_k \rangle^{1/2} \leq M$ (a priori valid from the moment equations under dissipation and bounded pumping), and the initial second-order connected correlations vanish. Write e for the difference between the exact first moments and the mean-field solution, and C for the second-order connected correlation. The linear GKSL part only damps C and does not source it; both the source of C and the drive of e carry a factor g ; under the moment bound $\dot{C} \leq 2M\|f\|gC + c_1 M^3\|f\|g$ and $\dot{e} \leq 2M\|f\|ge + \|f\|gC$, and two Grönwall iterations give

$$\text{err}(t) \leq \frac{c_1}{2} M^3 \|f\|^2 (gt)^2 e^{2M\|f\|gt}, \quad (69)$$

where c_1 is the constant absorbing the third-order connected-moment closure (assuming the truncated state has uniformly bounded moments to fourth order; c_1 grows with mode count and that moment bound): the self-similarity in the combined variable gt is precisely the shape of this bound, damping only lowering constants without entering the main term. Three-mode numerics check the same form: as $gt \rightarrow 0$, $\text{err}/(gt)^2$ tends to a constant (deviation $< 4\%$ for $gt \leq 0.33$), and the self-similar ratio $\text{err}(g/2, 2t)/\text{err}(g, t) \in [0.85, 0.98]$ ($gt \leq 1.2$). The constants in the bound depend on the in-window moment bound M , so the applicability of the mean-field equations is ultimately given by a priori moment control.

This appendix closes with the representation-theoretic standing of the interface F . For an odd prime q , the machine's phase classes contain all position functions, in particular the quadratic phases $e^{2\pi i b x^2/q}$; the group projectively generated by these together with F is exactly the Weil representation of $\text{SL}_2(\mathbb{F}_q)$ ($q = 7$ numerics: the relations $(ST)^3 \propto S^2$, $S^4 \propto I$, $T^7 \propto I$ verified, and the phase-invariant closure enumerates to order exactly $336 = |\text{SL}_2(\mathbb{F}_7)|$). F itself is uniquely determined by the Stone–von Neumann theorem on the Heisenberg pair of position functions and translations; the affine group $\text{Aff}(\mathbb{F}_q)$ is the Borel subgroup of $\text{PGL}_2(\mathbb{F}_q)$ fixing the point at infinity, and adjoining F is the Bruhat completion.

For odd primes q , the generating relations of the standard Weil representation hold at the full-ring layer at every truncation depth. On $L^2(\mathbb{Z}/q^K\mathbb{Z})$, position shifts and modulations generate the finite Heisenberg group; for each unitary in the operation set, conjugation preserves the Heisenberg group (up to phase), and the exponent pairs read off give $\text{SL}_2(\mathbb{Z}/q^K\mathbb{Z})$ matrices: the quadratic phase gives the unipotent $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, F the Weyl element $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, and the scaling M_u the torus element $\text{diag}(u, u^{-1})$. The standard generating relations yield all of $\text{SL}_2(\mathbb{Z}/q^K\mathbb{Z})$. Two finite levels are additionally checked at $q = 7$: the closure order at $K = 2$ is $115248 = |\text{SL}_2(\mathbb{Z}/49)|$, and at $K = 3$ it is $39530064 = |\text{SL}_2(\mathbb{Z}/343)|$, both exactly equal. By Stone–von Neumann uniqueness, the projective representation on the state space accordingly realizes the full $\text{SL}_2(\mathbb{Z}/q^K\mathbb{Z})$ Weil representation. In the chosen Weil normalization, F^2 projectively realizes the inversion induced by the central element $-I$. The listed discrete operations therefore sit inside one rank-two group action, the same group organizing the branching geometry (62) and the interface relation (37). This representation-theoretic closure constitutes no claim of universal

quantum computation, BQP power, fault tolerance, or complexity advantage.

C Certificate Systems and the Territory of Provability

This appendix gives the complete derivations of the two certificate systems, sieve and modular flow, and of their separation; all steps are self-contained and independently checkable. The main-text statement is §7.

C.1 The Sieve System and Parity Independence

The logical form of the sieve method is a certificate system. Fix $x \geq 2$: models are finitely supported nonnegative weights $w : [1, x] \cap \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$; axioms are the Type I data $A_d : |S_d(w) - g(d)X| \leq r_d$ ($S_d(w) = \sum_{d|n} w(n)$, $d \leq D$ squarefree, error budget $\sum r_d \leq R$); an upper-bound certificate is a finite vector (λ_d) satisfying the pointwise domination $\sum_{d|n} \lambda_d \geq \Phi(n)$, which yields for every model $\sum_n w(n)\Phi(n) \leq X \sum \lambda_d g(d) + \sum |\lambda_d| r_d$; lower-bound certificates are the reverse. Pointwise domination is the only inference rule, and soundness follows directly from $w \geq 0$. All the system knows about w is the data $\{S_d(w)\}$: this is the formalization of “Type I information only”.

Step 1, completeness. The provable bound equals the supremum over models: for a target $\Phi \geq 0$, the infimum over the certificate side equals the supremum over the model side (finite-dimensional linear-programming strong duality; the Slater condition is guaranteed by a strictly feasible model). Hence “unprovable” is equivalent to “a countermodel exists”: independence proofs are countermodel constructions.

Step 2, parity independence. Take the Liouville function λ and the two models $w_{\pm}(n) = (1 \pm \lambda(n))\mathbf{1}_{(\sqrt{x}, x]}(n)$. Complete multiplicativity gives $\sum_{d|n, n \leq y} \lambda(n) = \lambda(d)L(y/d)$ (L the partial sum of λ), and the prime number theorem’s cancellation estimate $|L(y)| \ll y \exp(-c\sqrt{\log y})$ puts both within the same axiom set $T(1/d, x - \sqrt{x}, x^{1-\varepsilon}, (r_d))$ ($T(g, X, D, (r_d))$ denoting the axiom set of density g , main-term scale X , modulus cap D , and error budget (r_d)); while the integers in $(\sqrt{x}, x]$ coprime to $P(\sqrt{x})$ are exactly the primes, so the sifted counts are 0 and $2\pi(x) + O(\sqrt{x})$ respectively ($\pi(x)$ the prime-counting function; the full computation is in §C.6). Conclusion:

Type I certificates give no positive lower bound for the prime count; upper bounds are off by a factor ≥ 2 .	(70)
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This is the independence form of the parity barrier [27]: one explicit pair of witnesses, same axioms, different targets. The root of the obstruction is model diversity: models need only be nonnegative, the multiplicative structure of n is forgotten, and the λ -twist and the counting measure are indistinguishable on Type I data. The algebraic identity of the sieve certificate cone: pointwise domination is precisely the commutative SOS cone of divisor convolutions (the Selberg weight $(\sum \lambda_d)^2$ is its typical element).

C.2 The Modular-Flow Certificate System

The axioms of the modular-flow system fall into two classes. K1 (algebra): the relations of the operation-set generators ($\mu_m \mu_n = \mu_{mn}$, $\mu_n^* \mu_n = 1$, $\sigma_t(\mu_n) = n^{it} \mu_n$), the algebraic embodiment of complete multiplicativity. K2 (state): $\varphi(1) = 1$, positivity $\varphi(a^* a) \geq 0$, σ_t -invariance, and the

KMS boundary condition; at the level of generator words the KMS condition is an algebraic constraint $(\sigma_{i\beta}(\mu_n) = n^{-\beta}\mu_n)$, involving no functional analysis. A proof in the modular-flow certificate system := a nonnegative linear combination of finite positivity instances $\varphi(a^*a) \geq 0$ with K1/K2 identities; the proof structure is isomorphic to the sieve system's LP certificates, only the certificate cone changes from the commutative divisor cone to the noncommutative *-SOS cone containing the automorphisms σ_t . Fairness: the two systems' certificates are both finite positive combinations, the same metamathematics, and the comparison is cone against cone; the sieve system is weak in its data interface (seeing only the diagonal commutative cone), not in mathematical resources.

The translation between the two systems: the restriction map res restricts a state φ to the spectral measure of the commutative subalgebra and takes the Type I functionals, giving the sieve shadow; sieve certificates pull back along res^* into the modular-flow certificate cone (commutative cone $\subset$ noncommutative cone), so the modular-flow system subsumes all reasoning of the sieve system. But the parity witness does not lift: the KMS state at $\beta \leq 1$ is unique (its proof needs only two elementary inputs, harmonic divergence and boundary convergence of Dirichlet series), the sieve shadow of the unique state is a product measure, whose λ -expectation $\sum \lambda(n)n^{-\beta}/\zeta(\beta) = \zeta(2\beta)/\zeta(\beta)^2$ tends to zero as $\beta \rightarrow 1^+$; whereas w_- has $\lambda \equiv -1$ on its support, with λ -expectation -1 .

C.3 Separation: Modular Flow Breaks the Parity Barrier

What the modular-flow certificate cone has extra is σ_t . Its power is exhibited by one positivity instance.

Step 1, take any element x and set $z = x + 2\sigma_t(x) + \sigma_{2t}(x)$ (coefficients 1, 2, 1, the binomial coefficients of $(1 + u)^2$). Expanding the single positivity axiom $\varphi(z^*z) \geq 0$ term by term: of the nine terms, the three diagonal ones merge by σ -invariance ($\varphi(\sigma_t a^* \sigma_t b) = \varphi(a^* b)$) into $(1 + 4 + 1)\varphi(x^*x)$; the cross terms pair into conjugates ($\varphi(\sigma_t a^* b) = \overline{\varphi(b^* \sigma_t a)}$), merging into $8 \text{Re } \varphi(x^* \sigma_t(x)) + 2 \text{Re } \varphi(x^* \sigma_{2t}(x))$; dividing by 2,

$$3\varphi(x^*x) + 4 \text{Re } \varphi(x^* \sigma_t(x)) + \text{Re } \varphi(x^* \sigma_{2t}(x)) \geq 0. \quad (71)$$

Step 2, insert the prime element $x_N = \sum_{p^k \leq N} k^{-1/2} p^{-k\sigma/2} \mu_{p^k}$ ($\sigma > 1$). Vanishing of the off-diagonal terms is a purely algebraic consequence of σ_t -invariance ($\varphi(a) = (n/m)^{it} \varphi(a)$ for all t forces $\varphi(a) = 0$), and the diagonal terms equal 1 by isometry; by the logarithmic expansion of the Euler product $\log \zeta(\sigma) = \sum_{p,k} k^{-1} p^{-k\sigma}$, as $N \rightarrow \infty$:

$$\varphi(x^*x) \rightarrow \log \zeta(\sigma), \quad \varphi(x^* \sigma_t(x)) \rightarrow \log \zeta(\sigma - it). \quad (72)$$

Substituting into (71) is precisely the Mertens inequality $3 \log \zeta(\sigma) + 4 \text{Re } \log \zeta(\sigma + it) + \text{Re } \log \zeta(\sigma + 2it) \geq 0$.

Step 3, if $\zeta(1 + it_0) = 0$ ($t_0 \neq 0$, of order $m \geq 1$), then as $\sigma \rightarrow 1^+$, $3 \log \zeta(\sigma) = 3 \log \frac{1}{\sigma-1} + O(1)$ and $4 \text{Re } \log \zeta(\sigma + it_0) = -4m \log \frac{1}{\sigma-1} + O(1)$; the third term, if $\zeta(1 + 2it_0) \neq 0$, is $O(1)$, and if it also vanishes, is negative and worse. Nonnegativity of the total forces $3 - 4m \geq 0$, contradicting $m \geq 1$. The boundary continuity needed comes from elementary partial summation ($\zeta(s) - \frac{1}{s-1}$ extends to $\text{Re } s > 0$), derivable within the system. Hence

$\zeta(1 + it) \neq 0 \quad (t \neq 0) \text{ is provable in the modular-flow certificate system.}$

(73)

Through a Tauberian theorem (Newman–Ikehara [29], the common analytic shell; the system’s net contribution is its entire input: nonnegativity, nonvanishing, continuation) the prime number theorem follows.

Step 4 (extension by the same template; the separation theorem needs only the first three steps), the arithmetic-progression version. With the character-weighted prime elements $x_{\chi,N} = \sum_{p^k \leq N} k^{-1/2} p^{-k\sigma/2} \chi(p^k) \mu_{p^k}$, take

$$z = x_N + 2\sigma_t(x_{\chi,N}) + \sigma_{2t}(x_{\chi^2,N}),$$

The same positivity expansion gives (cross terms simplified one by one, e.g., $\varphi(x_{\chi}^* \sigma_t(x_{\chi^2})) = \sum_{p,k} k^{-1} (\bar{\chi} \chi^2)(p^k) p^{-k(\sigma-it)} = \log L(\sigma - it, \chi)$, using $\bar{\chi} \chi^2 = \chi$; the χ -weighted diagonal terms lack the Euler factors at $p \mid q$, a deficit in the direction favorable to the inequality) $3 \log \zeta(\sigma) + 4 \operatorname{Re} \log L(\sigma + it, \chi) + \operatorname{Re} \log L(\sigma + 2it, \chi^2) \geq 0$: the Landau version of the inequality reappears verbatim, and the certificate identity is still a single positivity instance. All cases other than the Landau point (all complex χ , and real χ with $t \neq 0$) give $L(1 + it, \chi) \neq 0$ by the same order contradiction. The remaining Landau point (real χ , $t = 0$) goes by convolution positivity: $(1 * \chi)(n) \geq 0$ and $(1 * \chi)(n^2) \geq 1$ are K1 identities plus pointwise verification, and $\log(\zeta L)(\sigma) = \|\xi\|^2$ is a norm identity with nonnegative coefficients (where

$$\xi = \sum_{p,k} \left(\frac{1 + \chi(p^k)}{k} \right)^{1/2} p^{-k\sigma/2} \mu_{p^k}$$

acts on the reference vector); closure uses Landau’s positive-coefficient singularity theorem (a complex-analytic shell, of the same standing as the Tauberian one). The arithmetic-progression version of the prime number theorem is thereby certified.

Meanwhile (70) says sieve certificates cannot reach counts of prime-number-theorem precision. The separation theorem holds, and the demarcation follows:

The Type I commutative certificate cone admits a parity countermodel;
adding σ_t is a strict strengthening verified here.

(74)

Equations (71)–(73) prove that the modular flow suffices, within this certificate system, to recover the time-dependent information invisible to the Type I interface; they do not prove that every method breaking the parity barrier must be noncommutative, nor that σ_t is the minimal extension. Commutative systems enlarged by Type II information may likewise cross the Type I barrier; the comparison of proof-theoretic strength between the modular-flow extension and Type II extensions is left for further study. The territory and boundary of provability are in §C.5.

C.4 Critical Scaling: Sieve Asymptotics as a Phase-Transition Boundary Layer

The asymptotic theory of the sieve acquires a phase-transition reading in the modular-flow system. The sieve projection A_z (no prime factor $\leq z$) lies in the commutative subalgebra, and its Gibbs expectation is the exact Euler product $\varphi_{\beta}(A_z) = \prod_{p \leq z} (1 - p^{-\beta})$. The whole difficulty of the true sieve is the coupled double limit ($z \rightarrow \infty$, $\beta \rightarrow 1^+$); in the scaling variable $s = (\beta - 1) \log z$ the coupling decouples:

$$\lim_{z \rightarrow \infty} e^{\gamma} \log z \varphi_{\beta}(A_z) = G(s) = \exp \left(\int_0^1 \frac{1 - e^{-su}}{u} du \right) = s e^{\gamma + E_1(s)}.$$

(75)

Here γ is Euler's constant and $E_1(s) = \int_s^\infty e^{-v} v^{-1} dv$ the exponential integral. The two end limits recover the classical theorems: $s \rightarrow 0$ gives $G \rightarrow 1$ (Mertens), $s \rightarrow \infty$ gives $G \sim s e^\gamma$, i.e., $\varphi \sim \beta - 1$ (the ζ pole); sketch of the derivation: $\log \varphi_\beta(A_z) = -\sum_{p \leq z} \sum_k k^{-1} p^{-k\beta}$; substituting $\beta = 1 + s/\log z$ and changing variable to $u = \log p/\log z$, the main term $\sum_{p \leq z} p^{-\beta} = \sum_{p \leq z} p^{-1} e^{-su}$ differs from $\sum_{p \leq z} p^{-1}$ by a quantity converging to $\int_0^1 (1 - e^{-su}) u^{-1} du$ (the local form of Mertens' theorem giving the density du/u); the $k \geq 2$ layers converge uniformly and combine with the Mertens constant into the factor $e^{-\gamma}/\log z$. Numerics: fifteen combinations $s \in \{0, \frac{1}{2}, 1, 2, 5\} \times z \in \{10^4, 10^5, 10^6\}$; the $1/\log z$ first-order correction coefficient is stable (1.22/1.23/1.24 at $s = 1$; the numerical method is the direct Euler product up to the given z , Dickman's ρ by Euler stepping of its delay differential equation, step 5×10^{-4} , cutoff $u = 35$, and the duality identity checked to 2×10^{-3} at $s \in \{0.5, 1, 2, 5\}$). The sieve dimension $\varkappa$ (each channel excluding $\varkappa$ residue classes) adds channelwise to give $G_\varkappa = G^\varkappa$: the sieve dimension is the exponent of the scaling function, an arithmetic instance of a critical exponent.

The duality identity plugs in the smooth side. The Laplace transform of the Dickman function ρ satisfies the classical formula $\hat{\rho}(s) = \exp(\gamma + \int_0^s \frac{e^{-v}-1}{v} dv)$, and multiplied with $G(s) = \exp(\int_0^s \frac{1-e^{-v}}{v} dv)$ the exponents cancel:

$$\boxed{G(s) \hat{\rho}(s) = e^\gamma.} \quad (76)$$

Structural reading: the Euler factors of the rough and the smooth sides are complementary, $\zeta(\beta) \prod_{p \leq z} (1 - p^{-\beta}) \times \prod_{p \leq z} (1 - p^{-\beta})^{-1} = \zeta(\beta)$ exactly for each β ; in the scaling limit the rough side gives G , the smooth side gives $\log z \hat{\rho}$, and e^γ is the partition constant of the complementary decomposition. The classical sieve asymptotics (the Dickman–de Bruijn–Buchstab family [4]) thereby plug wholesale into the critical-scaling frame of the modular-flow system: sieve asymptotics may be read as the boundary-layer expansion at the $\beta = 1$ critical point of the ideal infinite BC system; the low-temperature stopping criteria of finite systems must be strictly distinguished from this infinite scaling limit.

C.5 The Territory of Provability

The system's provable range splits into two levels: certification completed (constructed and verified item by item, like (71) and its arithmetic-progression version) and candidate roadmap (main identities identified, but uniform quantifiers, constants, and analytic shells not yet assembled line by line).

First, the discrete Parseval identity

$$\sum_{b \bmod q} \left| \sum_n a_n e^{2\pi i b n / q} \right|^2 = q \sum_{n \equiv n'} a_n \bar{a}_{n'}$$

is the expansion of F -unitarity, a K1/K2 identity; it gives the finite positivity interface of the discrete circle method. But passing from fixed-modulus identities to major/minor-arc estimates uniform in the main parameter still requires parametrized certificate families.

Second, ternary Vinogradov and Bombieri–Vinogradov belong to the candidate roadmap, their gaps itemized as follows.

- (1) Vaughan's identity [28]: a K1 convolution identity, certificate written out (below);

- (2) Cauchy–Schwarz for the Type II bilinear blocks: a K2 positivity instance, identified; the certificate family uniform in the main parameter not yet constructed;
- (3) the large sieve inequality: the dual positivity of F -unitarity; the fixed-modulus case is the discrete Parseval above; in the uniformization over modulus families the duality step is K2 positivity and the Farey spacing count is elementary input, not yet assembled line by line;
- (4) the Siegel–Walfisz input on the major arcs: the Landau-point positivity certificate for real characters is given by §C.3, Step 4, but potential Siegel zeros make the explicit constants ineffective; this item is a structural obstruction to explicit constants in the certificate family, not an assembly gap;
- (5) the minor-arc bounds: Weyl differencing and completed sums are expressible as composites of pointwise domination with completed-sum identities; the line-by-line certificate is unwritten.

Items (2), (3), and (5) are workload gaps, and item (4) is an effectivity obstruction; until they are resolved or bypassed, the two theorems remain candidates, and a zero analytic shell cannot be asserted.

The certificate for item (1) is as follows. Write $\mu_{\leq V}$, $\mu_{>V}$ for the split of the Möbius coefficients by support at parameter V , and $\Lambda_{\leq U}$, $\Lambda_{>U}$ likewise. From the K1-layer identities $1 * \Lambda = \log$ and $\mu * 1 = \delta$:

$$(\mu_{>V} * 1) * \Lambda_{>U} = \Lambda_{>U} - (\mu_{\leq V} * 1) * \Lambda_{>U} = \Lambda_{>U} - \mu_{\leq V} * \log + (\mu_{\leq V} * \Lambda_{\leq U}) * 1,$$

and for $n > U$ one has $\Lambda_{>U}(n) = \Lambda(n)$, so

$$\Lambda(n) = [\mu_{\leq V} * \log](n) - [(\mu_{\leq V} * \Lambda_{\leq U}) * 1](n) + [(\mu_{>V} * 1) * \Lambda_{>U}](n) \quad (n > U). \quad (77)$$

The three terms are the log-weight Type I, the bounded-coefficient Type I, and the bilinear Type II block; every step is a K1 convolution identity with coefficient-support truncation, containing no analytic input ($U = 25$, $V = 40$, $n \leq 6000$ verified numerically term by term with zero violations). What remains in the gap ledger under items (2) and (5) is turning the summation ranges on the two sides of this identity into estimate certificates uniformly in the main parameter.

Third, the currently completed items are $\zeta(1+it) \neq 0$ (in-system, no shell), the prime number theorem (Tauberian shell), and the arithmetic-progression version of §C.3, Step 4 (Landau shell). Zero-density estimates and binary shifted convolutions remain candidates; the latter is expressible in the object language, but its estimative unreachability has as yet no certificate-complexity lower bound.

Fourth, the nature of the boundary. The expressive power of V_c and μ covers all objects of classical analytic number theory; the system’s boundary lies at the level of provability, not of expressibility.

C.6 Complete Proofs for the Sieve System

This subsection proves the three sieve-system conclusions of §C.1.

First, completeness (LP duality). Fix the theory $T = T(g, X, D, (r_d))$ and a target $\Phi \geq 0$, and suppose T has a strictly feasible model; the infimum of upper-certificate cost equals the supremum of the target over the model side. The model side is a finite-dimensional linear

program in the variables $w(n) \geq 0$ ($n \leq x$), with the two one-sided inequalities of each A_d as constraints. The dual assigns nonnegative multipliers $\lambda_d^\pm$ to the one-sided constraints and multipliers to the positivity constraints; eliminating the latter yields exactly the pointwise-domination form of the certificates, $\lambda_d = \lambda_d^+ - \lambda_d^-$, with cost exactly $X \sum \lambda_d g(d) + \sum |\lambda_d| r_d$. Strict feasibility gives Slater's condition, strong duality holds, and the certificate side attains. The lower-bound direction is identical. Hence “unprovable” is equivalent to “a countermodel exists”.

Second, the two-model lemma. Fix $\varepsilon \in (0, \frac{1}{2})$ and $w_\pm(n) = (1 \pm \lambda(n)) \mathbf{1}_{(\sqrt{x}, x]}(n)$. For squarefree $d \leq x^{1-\varepsilon}$, complete multiplicativity gives $\sum_{d|n, n \leq y} \lambda(n) = \lambda(d) L(y/d)$ ($L(y) = \sum_{m \leq y} \lambda(m)$); applying at $y = x$ and $y = \sqrt{x}$, divisor counting gives the main term:

$$S_d(w_\pm) = \frac{x - \sqrt{x}}{d} + O(1) \pm \lambda(d) \left(L(x/d) - L(\sqrt{x}/d) \right).$$

The prime number theorem's cancellation $|L(y)| \ll y \exp(-c\sqrt{\log y})$ at $y = x/d \geq x^\varepsilon$ gives $r_d \ll 1 + (x/d) \exp(-c\sqrt{\varepsilon \log x})$; summing over $d \leq D$ costs a $\log x$ factor, absorbed by lowering the exponent constant, giving $R \ll x \exp(-c'\sqrt{\log x})$. So $w_\pm$ jointly satisfy $T(1/d, x - \sqrt{x}, x^{1-\varepsilon}, (r_d))$.

Third, the parity theorem. $\Phi_z(n) = \mathbf{1}_{\gcd(n, P(z))=1}$, $z = \sqrt{x}$. Integers in $(\sqrt{x}, x]$ coprime to $P(\sqrt{x})$ have all prime factors exceeding $\sqrt{x}$, and two such factors would exceed x , so they are exactly the primes:

$$\sum_n w_\pm(n) \Phi_{\sqrt{x}}(n) = \sum_{\sqrt{x} < p \leq x} (1 \pm \lambda(p)) = \sum_{\sqrt{x} < p \leq x} (1 \mp 1),$$

equal to 0 for w_+ and $2(\pi(x) - \pi(\sqrt{x}))$ for w_- . Both are models of T (the second step), and by the first step the provable bounds are pinned to the window $[0, 2\pi(x) + O(\sqrt{x})]$: no lower-bound certificate gives a positive lower bound; no upper-bound certificate goes below $(2 - o(1))\pi(x)$, even though the counting measure $w_0 = \mathbf{1}_{(\sqrt{x}, x]}$ (also a model of T) satisfies $\sum w_0 \Phi = \pi(x) - \pi(\sqrt{x}) + O(1)$.

Remark (non-circularity): the witness construction uses the prime number theorem at the meta-level, a fact about the actual integer sequence; the independence theorem is a statement about the reasoning power of the certificate system. §C.3 derives the prime number theorem inside the modular-flow system, and no circle arises: witnesses belong to the model space outside the system, certificates to the certificate cone inside it.

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